

Mathematical Supplement

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The domain convention in main Section 2.0 applies to the point-mass formulas below. The measurable-event hypotheses there also apply throughout; S19 proves sufficient regularity and specifies the capacity, composition, error and statistical uses.

Article: *Causal Foundations and Experimental Certification of Contingent Assignment, Heredity, and Reflexive Maintenance*

This supplement develops the arguments behind the main article. Part I gives countermodels, proof details, finite-sample arguments, and condition dictionaries. Part II states the certificate for one-step maintenance of actual future assignment capacity. Part III constructs a finite model satisfying the joint conditions.

The abbreviations used below include OA (operational assignment), SCM (structural causal model), TV (total variation), and IID (independent and identically distributed).

Section numbers restart within Parts II and III; references to a numbered section within those parts refer to that same part unless another part is named. The main article provides the assignment assay definition and mathematical results to which these constructions apply.

Part I. Countermodels, proof details, and finite-sample arguments

S1. Decoder-free provenance is not passively identifiable

S1.1 Passive observational witness pair

Let I_t be an externally specified input process, Y_t the observable output process, and Θ a finite assignment organization.

Fix:

- a distribution ν on Θ ;
- a pre-activation observable kernel $K_{\perp}$;
- post-activation observable kernels K_{θ} .

Construct two models.

Emergent model E

At the origin there is no assignment-bearing organization. Until an activation time τ ,

$$Y_t \sim K_{\perp}(\cdot \mid I_{\leq t}).$$

At τ , sample

$$\Theta_E \sim \nu$$

and thereafter use

$$Y_t \sim K_{\Theta_E}(\cdot \mid I_{\leq t}).$$

Preinstalled-dormant model P

At the origin sample

$$\Theta_P \sim \nu$$

and store it in a dormant internal variable. Before τ , the observable mechanism ignores it and again uses $K_{\perp}$. At τ , the stored value becomes active and the observable mechanism uses

$$Y_t \sim K_{\Theta_P}(\cdot \mid I_{\leq t}).$$

Equality of passive laws

The two models have:

1. identical pre- τ observable laws;

2. the same distribution ν of the active organization;
3. the same post- τ kernel conditional on that organization.

Therefore their complete passive observable path measures are identical:

$$\boxed{\mathcal{L}_E(I_{0:T}, Y_{0:T}) = \mathcal{L}_P(I_{0:T}, Y_{0:T}) \quad \forall T.}$$

Yet their origin provenance differs: E has no assignment organization at the origin, whereas P does.

Hence any statistic measurable with respect to the passive input/output path sigma-field has the same law in both models. Decoder-free provenance is therefore not identifiable on a model class containing both witnesses.

This is an identifiability no-go, not a claim that passive data are always scientifically useless.

S1.2 Explicit origin-audit labels for the same-law witnesses

To bind this no-go to the origin predicate in main Section 2.4, specialize Θ to the two physical matchings M_0, M_1 , with uniform ν , and use a common output gate. At the origin E has empty matching and P has stored matching M_Θ ; both gates are closed. At τ , E creates a uniform matching, while P retains its stored matching, and the gate opens. This gives the same passive and input-only transcript laws proved above.

Use the same predeclared origin audit for both systems: disable new link formation during the diagnostic word, allow a common gate opening, any declared edge-set deletion, and either input challenge. Include the whole orbit of every such word under $S_2(U) \times S_2(V)$. These operations never create a link or select a target matching. A delayed read alone, if it permits new link formation, would not be an origin-preserving diagnostic.

For each realized P origin, gate opening without deletion exposes its actual matching channel; deletion of its links removes the response. Each stored nonempty matching is thus diagnostically exposed. The E origin gives only null channels, since it has no link and origin diagnostics disable their formation. DC0 holds on this declared matching ontology, and the two full origin verdicts differ. Averaging the two stored P channels before applying the diagnostic would lose this state-indexed information and is not the origin criterion.

The restricted observational/input-only interface in S1-S2 does not include these origin diagnostic operations. Thus the different audit labels and identical restricted transcript laws coexist. If the gate-opening diagnostic is removed, deletability alone leaves the dormant example inconclusive under DC0; it does not turn it into an empty-origin system.

S2. Input-only interventions do not universally identify provenance

Use the same two models E, P . Allow an experimenter to choose the input I_t arbitrarily, including adaptively as a function of the previously observed transcript.

The crucial condition is that the experimenter may intervene only on the declared input process and cannot inspect, set, condition on, or intervene on the hidden assignment-bearing organization.

For every observable history h_t and every chosen input i_t , the controlled next-observable kernel is the same in E and P :

$$P_E(Y_{t+1} \in \cdot \mid h_t, do(I_t = i_t)) = P_P(Y_{t+1} \in \cdot \mid h_t, do(I_t = i_t)).$$

The result follows by induction on the transcript length.

If an adaptive policy $\pi_t(di \mid h_t)$ selects the next input from the same observed history in both models, then

$$\begin{aligned} P_E(H_{t+1} \in dh', I_t \in di) \\ = P_E(H_t \in dh) \pi_t(di \mid h) K(dy \mid h, i), \end{aligned}$$

and exactly the same recursion holds under P . Starting from the same initial observable law gives

$$\boxed{\mathcal{L}_E(H_T) = \mathcal{L}_P(H_T)}$$

for every finite adaptive input policy.

Thus arbitrary open-loop or adaptive control of **inputs alone** cannot universally identify the hidden provenance distinction in this witness class.

The result ceases to apply once the intervention/measurement family directly accesses provenance-bearing internal state.

S3. DFI-A: representative deletion and structural-boundary countermodels

The current relation-relative certificate has the form

$$M1 + M3^\dagger + C0 + C1^* + C2^* \implies \text{basic relation-relative CLAIM-A.}$$

The following finite models identify the assumptions needed for this sufficient criterion.

S3.1 Delete C0

Let the origin state be

$$A_0 = a_0 \in \mathcal{A}_+$$

almost surely. Choose two later candidate states with different matched structural response channels and make the certification regimes satisfy $C1^*, C2^*$.

Then all later channel conditions can hold, but the origin-null requirement is false. Hence C0 is deletion-indispensable for this certificate architecture.

S3.2 Delete C1*

Let

$$P(A_0 = \perp) = 1$$

and define a structural kernel independent of candidate assignment state:

$$K(v \mid u, a, d) = K_0(v \mid u, d).$$

Let the two certification regimes have the same nuisance law. Then

$$\mathcal{K}_{a_0, \mu} = \mathcal{K}_{a_1, \mu}.$$

No contingent A -dependent structural response-channel family exists. Parent/later state labels alone do not create the certificate.

S3.3 Delete C2*: unmatched nuisance false positive

Take $\mathcal{U} = \mathcal{V} = \mathcal{D} = \{0, 1\}$, $\mathcal{A} = \{\perp, a_0, a_1\}$, $\mathcal{A}_+ = \{a_0, a_1\}$, and $u^* = 1$. The declared internal organization is exhausted by A ; D is an external nuisance coordinate. Set the origin and the unmanipulated baseline organization to $\perp$. Use the total acyclic structural equations

$$D = d(A), \quad d(\perp) = 0, \quad d(a_i) = i, \quad V = U \oplus D.$$

Output noise is a point mass. The equations are fixed across the ideal joint surgeries $J_i(u) = do(A = a_i, U = u)$; neither equation contains a protocol label. Under $J_i(1)$, $D = i$ and $V = 1 \oplus i$, so the observed output laws have total-variation distance one. The two nuisance laws are δ_0 and δ_1 , violating $C2^*$. Nevertheless, for every common nuisance law μ ,

$$\mathcal{K}_{a_0, \mu} = \mathcal{K}_{a_1, \mu},$$

because the stable output kernel $K(v \mid u, a, d)$ does not depend on a . This supplies a total realization of the unmatched-nuisance scheme under the prescribed organization/input interventions. Within this declared ontology the origin and mechanism/interface conditions remain in force; they make no additional claim that the fixed law itself is decoder-free. The apparent assignment-state effect is entirely an unmatched-nuisance effect.

S3.4 Remove M1: hidden decoder outside the ontology

Introduce a latent origin-time decoder variable Z outside the declared A -ontology. Let

$$A_0 = \perp$$

inside the declared ontology and satisfy the later certification conditions.

The formal later channel comparison can still pass. Absence of a decoder-bearing organization inside the declared A -ontology remains true; what fails is the stronger absence claim covering all relevant internal organization. The omitted Z makes that ontology nonexhaustive, so M1 is false. This example cannot refute the full theorem with M1 in force. The two absence targets must not be interchanged.

Thus $M1$ is a model-class/semantic boundary, not an empirically inferred fact.

S3.5 Protocol aliasing: a structural boundary for $M3^\dagger$

Let J set the apparent state $A = a_i$, keep nuisance fixed, but let the actual output depend directly on the protocol:

$$V = \begin{cases} 0, & J = j_0, \\ 1, & J = j_1. \end{cases}$$

Then the observed certification outputs differ even if the declared A -indexed structural kernel is A -insensitive.

An omitted path

$$J \rightarrow H \rightarrow V$$

has the same defect.

This is why the structural parent-set and intervention-stability conditions in $M3^\dagger$ are load-bearing. Here J labels the actual protocol; its effects are not identified with the literal joint surgery $do(A = a_i, U = u)$. A surgery on A, U alone does not change a separate protocol parent J . The example therefore establishes a protocol-aliasing/structural-class boundary, not a single-block deletion witness retaining all the other literal-surgery premises. The deletion claims for $C0, C1^*, C2^*$ are the separate constructions in S3.1-S3.3.

Scope

These countermodels show deletion indispensability or model-class dependence **within this certificate architecture**. They do not prove that the exact blocks are globally necessary.

S4. DFI-H: representative heredity countermodels

The aligned distributional heredity theorem uses

$$H0^* + H1^* + H23^* + H5^\dagger.$$

S4.0 Complete finite-state heredity dictionary and proof

Let P, L, C be finite parent, lineage and child assignment-relevant states, and let W be the complete finite child context. Fix distinct p_0, p_1 , distinct l_0, l_1 , and an event B in the child state space. The surgical arms are $G_i = do(P = p_i)$ and $I_i = do(L = l_i)$, for $i = 0, 1$; no other equation is directly replaced.

H5 $\dagger$.1: $C = F_H(L, W, E_H)$, with no endogenous parent of C outside (L, W) . **H5 $\dagger$.2:** E_H is independent of the complete upstream exogenous vector generating P, L, W . **H5 $\dagger$.2a:** under each of the four arms, $E_H \sim \lambda_H$ and $E_H \perp W$. **H5 $\dagger$.3:** F_H and λ_H are unchanged across these arms. These requirements are part of the structural hypothesis, not consequences of matching noise marginals.

Define

$$Q(c \mid l, w) = P_{\lambda_H}(F_H(l, w, E_H) = c), \quad Q_{l, \mu}(c) = \sum_w Q(c \mid l, w) \mu(w).$$

H0*: $P(L = l_i \mid G_i) = 1$. **H1*:** $P(C \in B \mid I_0) \neq P(C \in B \mid I_1)$. **H23*:** $\mathcal{L}(W \mid G_i) = \mathcal{L}(W \mid I_i) = \mu$ for both i , matching the full joint context, not just its component marginals.

Basic **CLAIM-H** means aligned parent-to-lineage loading, distinct context-averaged child-response kernels for the aligned lineage states, distinct child-event probabilities under the corresponding parent arms, and the exhaustive child interface (L, W) . It does not assert state equality or exclusive, unit-level mediation through L . A parent effect through W that preserves its regime-wise law is not excluded.

Proof of Theorem 8. Under either G_i or I_i , the lineage value is l_i almost surely, and the joint law of (W, E_H) is $\mu \otimes \lambda_H$. The same structural function therefore gives, for every child event D ,

$$P(C \in D \mid G_i) = Q_{l_i, \mu}(D) = P(C \in D \mid I_i).$$

H1 makes the two values on B distinct. H0 and H5[†].1 supply the other clauses of basic CLAIM-H. Thus $\Delta_H = d_{TV}(\mathcal{Q}_{l_0,\mu}, \mathcal{Q}_{l_1,\mu}) > 0$. This is the finite theorem used here. Integral versions require the corresponding measurable structural kernels and hypotheses, rather than being inferred from notation alone. $\square$

The importance of H5[†].2a can be seen with a fair bit E and equations $P = 0, L = P, W = P \oplus E, C = L \oplus W \oplus E$. The four arms preserve the separate fair marginals of W and E, but alter their coupling. Lineage arms have child contrast 1 while parent arms have contrast 0. This model violates H5[†].2a (and the stated upstream-noise independence); it is not a counterexample under the full dictionary.

S4.1 Delete H0*

Let context be constant and define

$$C = L.$$

The lineage interventions

$$do(L = 0), \quad do(L = 1)$$

produce different child laws, so H1* holds.

But let both parent interventions leave

$$L = 0$$

almost surely. Then

$$P(C \mid G_0) = P(C \mid G_1).$$

The lineage carrier is causally relevant, but the parent state does not load it. There is no parent-to-child heredity.

S4.2 Delete H1*

Let parent interventions load distinct lineage states $L = l_i$ as required by H0*. Keep context constant but define

$$C = c_0$$

for every lineage state.

Then parent-to-lineage loading exists but

$$\mathcal{Q}_{l_0,\mu} = \mathcal{Q}_{l_1,\mu}.$$

No child effect exists.

S4.3 Delete joint context matching H23*

Take binary P, L, E, C , constant $R = r_0$, with baseline equations

$$P := 0, \quad L := P, \quad E := P,$$

and child mechanism

$$C := L \oplus E.$$

Under lineage intervention $I_i = do(L = i)$, the baseline parent remains $P = 0$, hence $E = 0$ and

$$C = i.$$

Thus H1* holds.

Under parent intervention $G_i = do(P = i)$,

$$L = i, \quad E = i,$$

so

$$C = i \oplus i = 0$$

under both parent regimes.

Hence H0*, H1*, H5[†] may all hold while the parent-to-child conclusion fails. The failure is entirely due to the unmatched context

$$W = (E, r_0).$$

S4.4 Remove $H5^\dagger$: bypass cancellation

Take constant context and

$$C = L \oplus P.$$

Under lineage intervention $do(L = i)$, keep baseline $P = 0$, yielding $C = i$.

Under parent intervention $do(P = i)$, let lineage loading give $L = i$, yielding

$$C = i \oplus i = 0.$$

Thus the lineage intervention looks positive while the parent intervention has no child effect because a direct parent bypass cancels the lineage contribution.

S5. DFI-R(loop): representative support-loop countermodels

The support-loop/re-entry theorem is evaluated at its stated support-response target.

S5.1 Delete $R1^*$: one-way support is insufficient

Use binary labels $a_i = s_i = i$, $i = 0, 1$, and let

$$S = A$$

under organization interventions, so the forward organization-to-support link is genuine.

Let

$$A^+ = a_0$$

for every support state.

Then

$$M_{s_0, \mu} = M_{s_1, \mu}.$$

A genuine forward support edge alone does not establish a maintenance advantage.

S5.2 Delete $R0^*$: no forward loading

Use constant context. Let

$$F_S(a_0) = s_0, \quad F_S(a_1) = s_0$$

so the two witness organizations have identical support laws. Add a state b with $F_S(b) = s_1$, and let maintenance map

$$s_0 \mapsto a_0, \quad s_1 \mapsto b.$$

A support intervention can still favor the support-loop re-entry class, but the designated organization pair does not load distinct support states. The forward link fails.

S5.3 Delete $R23^*$: context shift fakes the aligned composition

Take binary A, S, W, A^+ with

$$A := 0, \quad S := A, \quad W := A,$$

and

$$A^+ := S \oplus W.$$

Use reference law $\mu = \delta_0$.

Under support intervention $I_i = do(S = i)$, baseline $A = 0$ leaves $W = 0$, so

$$A^+ = i.$$

The support intervention can therefore appear maintenance-positive.

Under organization intervention $G_i = do(A = i)$,

$$S = i, \quad W = i,$$

hence

$$A^+ = i \oplus i = 0$$

under both regimes.

The mismatch of W destroys the aligned composition.

S5.4 Support response need not imply assignment capacity

The support-response forward-active class is

$$\mathcal{A}_\circ = \{a : \Sigma_a \neq \Sigma_{a_0}\},$$

which is a support-response class, not an operational-assignment-capacity class.

Let

$$\mathcal{A} = \{a_0, a_1, b, c\}.$$

Here the capacity class is derived from a fixed operational assay. Use challenges $u \in \{0, 1\}$, responses $\{0, 1, \emptyset\}$, and

$$\text{read}(a_0, u) = u, \quad \text{read}(a_1, u) = 1 - u, \quad \text{read}(b, u) = \text{read}(c, u) = \emptyset.$$

Admit whole-organization ablation to c , followed by either challenge. A two-element symmetry group swaps a_0, a_1 and the binary response symbols, fixing b, c , both challenge labels, and the null response. The assay is equivariant. States a_0, a_1 have distinct challenged responses, an ablation-sensitive channel, and a nontrivial channel orbit; b, c have only the invariant null channel. Consequently the operational-assignment capacity class on this four-state domain is exactly

$$\mathcal{A}_{\text{cap}} = \{a_0, a_1\}.$$

At the future assay cut take $C^+ = A^+$ and use this same assay, with no hidden additional state coordinate. The following support/maintenance equations do not consult the assay or its capacity classification.

Use

$$a_0 \mapsto s_0, \quad a_1 \mapsto s_1, \quad b \mapsto s_1, \quad c \mapsto s_0,$$

and

$$s_0 \mapsto c, \quad s_1 \mapsto b.$$

Then

$$\mathcal{A}_\circ = \{a_1, b\}$$

and the support-loop re-entry advantage is positive, while

$$P(A^+ \in \mathcal{A}_{\text{cap}} \mid I_0) = P(A^+ \in \mathcal{A}_{\text{cap}} \mid I_1) = 0.$$

Thus DFI-R(loop) support-loop certification does not imply actual future operational-assignment capacity.

The capacity-guarded certificate in Part II supplies the additional bridge and binds the later organization state to the actual future interventional signature.

Common re-entry completion for S5.1–S5.4. These schemes can all use point-mass support and maintenance disturbances and one fixed deterministic function for each of the two printed mechanisms. Use baseline $A = a_0$ (binary 0 in S5.1 and S5.3). In S5.1, S5.2, and S5.4 take a singleton context with reference law μ ; in S5.3 retain $W = A$ and $\mu = \delta_0$. Predeclare a re-entry regime which, at this baseline, clamps $S = s_{B_{\text{re}}}$, where $B_{\text{re}} \sim \text{Bernoulli}(1/2)$ is independent, and then uses the unchanged maintenance equation. The resulting laws of A^+ are

scheme	$\mathcal{A}_{\text{cert}}$	$\mathcal{L}_R(A^+)$
S5.1	$\{a_0\}$	δ_{a_0}
S5.2	$\{a_0, b\}$	$(\delta_{a_0} + \delta_b)/2$
S5.3	$\{0, 1\}$	$(\delta_0 + \delta_1)/2$
S5.4	$\{b, c\}$	$(\delta_b + \delta_c)/2$

For every covered $A^+ = a$, set W_{next} to the reference context, take $E_{S, \text{next}} = 0$, and use the same F_S at the next support step. This gives the required conditional context and disturbance laws and

$$\mathcal{L}_R(S_{\text{next}} \mid A^+ = a) = \Sigma_{a, \mu}.$$

Thus R6† is satisfied, including full certification-support coverage. With these explicit completions S5.1 violates R1* only, S5.2 violates R0* only, and S5.3 violates R23* only among the S16 support-loop blocks.

S5.4 satisfies all those blocks and separates their support-response target from the actual future OA target of the fixed assay. No second maintenance cycle or capacity conclusion is inferred from R6†.

S6. UCCE: direct nonfactorization and near-miss countermodels

The first two examples distinguish baseline state binding from conditional continuation laws, and context matching from kernel selection. The next two show why pairwise consistency need not yield a common probability law or a common acyclic causal model. S17 states the full sufficient composition conditions; Part III gives a model satisfying them.

S6.1 Baseline binding does not determine boundary continuation

Let $Q \sim \text{Bernoulli}(1/2)$. In a finite acyclic structural model set

$$X_A = Q, \quad V_A = X_A, \quad P_H = Q, \quad L_H = P_H, \quad C_H = L_H, \quad X_R = C_H, \\ S_R = X_R \oplus W_R, \quad A_R^+ = S_R.$$

The baseline states satisfy $X_A = P_H = L_H = C_H = X_R = Q$ pointwise. Under the standardized context intervention $do(W_R = 0)$, the continuation at the H/R boundary is $(S_R, A_R^+) = (q, q)$ conditional on $X_R = q$. Under the actual context $W_R = 1$, it is $(1 - q, 1 - q)$. For each $q \in \{0, 1\}$, these two conditional laws have disjoint singleton supports and total-variation distance one. Their unconditioned laws, after mixing over the fair bit, are equal; the distinction is explicitly a bound-state conditional one.

Thus

$\text{baseline state binding} \not\Rightarrow \text{conditional boundary continuation matching.}$

This is the full strength asserted for this example. Under the surgical intervention $do(X_A = i)$, $V_A = i$ but $P_H = Q$ is unchanged; A/H binding then holds with probability $1/2$, not one. The model therefore does not satisfy the full regime-wise binding requirement in S17. Local response differences in this example do not establish a complete R(loop) certificate, and no claim of complete local A/H/R PASS is used in the conclusion. S17 proves a sufficient composition theorem with its full premises; Part III supplies a separate positive construction satisfying them.

S6.2 context matching alone does not fix kernel selection

Let

$$Q \sim \text{Bernoulli}(1/2), \quad S := Q, \quad \Theta := S, \quad W := 0,$$

and

$$Y := \Theta \oplus Q.$$

In the fixed target standard regime,

$$\Theta = Q$$

so

$$Y = 0$$

almost surely, and the fixed conditional response satisfies

$$g(\theta, 0) = 1$$

for the event $Y = 0$.

Now use the source-faithful intervention

$$do(S = \theta).$$

State binding and literal context matching still pass:

$$\Theta = \theta, \quad W = 0.$$

But the intervention breaks the correlation between Θ and Q , giving

$$P(Y = 0) = \frac{1}{2}.$$

Therefore the actual boundary continuation kernel differs from the fixed conditional kernel.

A context condition alone cannot replace B_{kernel} .

S6.3 pairwise probability gluing can fail globally

The following finite probability example illustrates an established local-to-global obstruction; see [31, Sections 2.5–3 and 7]. Its application to the specified local certificates is given below.

Let binary X, Y, Z have three proposed pair laws satisfying

$$\begin{aligned} X \oplus Y &= 1, \\ Y \oplus Z &= 1, \\ Z \oplus X &= 1 \end{aligned}$$

almost surely, with each allowed pair equally likely.

Every singleton overlap marginal is Bernoulli(1/2), so the pair laws are pairwise consistent on overlaps.

But XORing the three required equalities gives

$$0 = 1.$$

No global joint law exists.

Thus

$$\boxed{\text{pairwise overlap-law consistency} \not\Rightarrow \text{global joint gluing.}}$$

Extension preserving complete local certificates. Take the valid local A, H, and R(loop) certificate modules of Part III separately. Append to their nuisance/context coordinates, respectively, the pairs (X, Y) , (Y, Z) , and (Z, X) with the above fair anticorrelated laws. Within each module the appended pair is independent of every original variable and disturbance. All original mechanisms ignore it, and every required local intervention preserves its law. The enlarged reference context is the product of the old reference context law and this pair law. Thus all original response kernels, contrasts, context equalities, surgical interfaces, and noise-separation conditions are retained. For R6† append a fresh independent pair with the same law to the next context; conditional on every covered A^+ , the next context and disturbance conditions are therefore retained as well. These are three valid local certificates.

Now prescribe that coordinates with the same names are identified across their overlaps. A common realization would have to satisfy the three displayed XOR equalities almost surely, which is impossible by the preceding calculation. The obstruction concerns this specified overlap datum. Without those identifications the independent modules can coexist; the original compatible Part III construction also remains available. This does not contradict the UCCE implication, whose full global compatibility hypotheses are not met, and does not strengthen the separate baseline-only example S6.1.

S6.4 pairwise acyclic causal charts can create a global cycle

Let

$$Q \sim \text{Bernoulli}(1/2).$$

Use three local charts:

$$\begin{aligned} X &:= Q, & Y &:= X; \\ Y &:= Q, & Z &:= Y; \\ Z &:= Q, & X &:= Z. \end{aligned}$$

Each chart is individually acyclic. Their passive pair laws are compatible with

$$X = Y = Z = Q.$$

Yet reproducing all three intervention effects in one global acyclic SCM would require

$$X \prec Y, \quad Y \prec Z, \quad Z \prec X,$$

which is impossible.

Thus pairwise causal-chart compatibility is insufficient for one global acyclic surgical semantics.

S6.5 Why UCCE is not merely target restatement

The key decomposed continuation conditions are:

1. actual boundary kernel invariance,

$$K^{\text{act}}(\theta, w) = g(\phi(\theta), w);$$

2. complete fixed-response-signature context equality,

$$\sum_w g(\phi(\theta), w) \nu_\theta(w) = \sum_w g(\phi(\theta), w) \mu_{\phi(\theta)}(w).$$

Together, by the actual law of total probability, they imply the fixed target continuation law.

These conditions are not identical to simply assuming the target continuation law:

- literal equality $\nu = \mu$ is **stronger** than necessary;
- a hand-picked partial response signature is **weaker** than sufficient;
- The countermodel in S6.2 shows that the context signature alone does not control intervention-induced kernel selection.

The scientific content is therefore the nonfactorization and the noncircular decomposition of cross-boundary obligations.

S7. UFE: structural-softening no-go

Let the intended protected mechanism be

$$C = L$$

for binary L, C .

Introduce a hidden binary variable Z and let the implemented global mechanism be

$$C = L \oplus Z.$$

Suppose every required certification regime happens to have

$$Z = 0.$$

Then every required-regime probability law of the implemented model agrees exactly with the intended model:

$$d_{\text{TV}}(P_{\text{required}}^{\text{impl}}, P_{\text{required}}^{\text{ref}}) = 0.$$

However, the allowed off-regime intervention

$$do(Z = 1)$$

reveals

$$C = 1 - L,$$

so the structural mechanism is wrong.

Therefore no small required-law TV defect, even zero defect, can replace the exact structural/surgical gate.

S8. UFE primitive error transport

Condition the model law on a required cell z of positive probability. Let G denote correct state binding and let T be the full joint tested continuation response. The cell and G are measurable events, T and W are measurable random variables, the conditional laws used below are measurable probability kernels, and the integrated TV defects are measurable. These are explicit hypotheses of a measurable extension (S19.3); finite discrete kernels satisfy them. Set $s_z = P(G^c \mid z)$. If $s_z < 1$, put

$$\nu_z^G = \mathcal{L}(W \mid z, G), \quad K_z^{\text{act}}(\cdot \mid w) = \mathcal{L}(T \mid z, G, W = w).$$

Let $K_z^{\text{ref}}(\cdot \mid w)$ be the intended continuation kernel and μ_z its fixed reference context. Define $R_z = \int K_z^{\text{ref}}(\cdot \mid w) \mu_z(dw)$, and

$$\begin{aligned}
k_z &= \int d_{\text{TV}}(K_z^{\text{act}}(\cdot | w), K_z^{\text{ref}}(\cdot | w)) \nu_z^G(dw), \\
c_z &= d_{\text{TV}}\left(\int K_z^{\text{ref}}(\cdot | w) \nu_z^G(dw), R_z\right), \\
\delta_z &= d_{\text{TV}}(\mathcal{L}(T | z), R_z).
\end{aligned}$$

Finite kernels suffice here; a measurable extension requires regular conditional kernels on the declared spaces. The two matched-branch errors use the same ν_z^G . Substituting the unconditional-on-G context $\mathcal{L}(W | z)$ is not licensed. Upper bounds on these defined errors may be used in the formula. When $s_z = 1$, use $\delta_z \leq 1$ without constructing a kernel on the empty matched branch.

Then the realized continuation defect obeys the decomposition

$$\boxed{\delta_z \leq s_z + (1 - s_z) \min\{1, k_z + c_z\}.}$$

Proof sketch

Split the realized continuation law according to whether the handoff state mismatches the target state.

The mismatch branch has mass s_z and contributes at most s_z in TV.

On the matched branch, compare:

1. the actual continuation kernel to the intended kernel under the actual context, costing at most k_z ;
2. the intended kernel averaged under actual versus fixed context, costing at most c_z .

By triangle inequality, the matched-branch defect is at most $\min\{1, k_z + c_z\}$. Weight by $1 - s_z$ and add the mismatch branch.

Hence the theorem derives the final continuation budget from lower-level state/kernel/context errors; it is not simply a restatement of the approximate target.

S9. Protocol-layer support amplification

If two joint laws P, Q satisfy

$$d_{\text{TV}}(P, Q) \leq \varepsilon$$

and an event E has reference support

$$Q(E) \geq h > 0,$$

then, provided also that $P(E) > 0$,

$$d_{\text{TV}}(P(\cdot | E), Q(\cdot | E))$$

is bounded above by

$$\min\left\{1, \frac{\varepsilon}{h}\right\}.$$

For completeness, let $p = P(E) > 0$, $q = Q(E) > 0$ and $B \subseteq E$. If $d = P(B)/p - Q(B)/q \geq 0$, then for $p \geq q$, $qd \leq P(B) - Q(B)$, and for $p \leq q$, $qd \leq Q(E \setminus B) - P(E \setminus B)$. Either is at most $d_{\text{TV}}(P, Q)$. A negative difference is handled by replacing B with its complement within E. Taking the supremum gives the printed constant-one bound. If $P(E) = 0$, the realized conditional quantity is unassessable; the reference floor alone does not define it.

The $1/h$ amplification is unavoidable in rare cells.

Conditioning can amplify total-variation error. UFE therefore tracks both transcript error and the support of each handoff cell.

S10. UFE realized-cell theorem

Let:

- s_z^{mod} be the model-layer state mismatch;
- k_z^{mod} the model-layer continuation-kernel defect;

- c_z^{mod} the response-weighted context defect;
- ι_z the protocol/transcript-law TV defect;
- $h_z > 0$ a fixed model-reference support floor for the required cell.

Use exactly the matched-branch definitions in S8. Let Q be the model law and P the realized protocol law on a common transcript that determines z , G and T , or has a common downstream kernel to their full joint record. Require $d_{\text{TV}}(P, Q) \leq \iota_z$, $Q(z) \geq h_z$ and $P(z) > 0$. TV contraction through the common kernel and S9 control the conditional joint record by $\min(1, \iota_z/h_z)$. Comparing only selected marginals or a transcript that cannot determine the required cell does not establish these hypotheses.

Define

$$\sigma_z = \min \left\{ 1, s_z^{\text{mod}} + \frac{\iota_z}{h_z} \right\},$$

and

$$\beta_z = \min \left\{ 1, \frac{\iota_z}{h_z} + s_z^{\text{mod}} + (1 - s_z^{\text{mod}}) \min\{1, k_z^{\text{mod}} + c_z^{\text{mod}}\} \right\}.$$

Then the realized state and continuation defects satisfy

$$\boxed{s_z^{\text{real}} \leq \sigma_z, \quad \delta_z^{\text{real}} \leq \beta_z.}$$

This closes the chain

primitive structural/protocol errors $\rightarrow$ realized handoff/continuation budgets.

S11. UFE local margin propagation

If the two implemented comparison-arm laws for an A/H/R diagnostic lie within TV distances η_0, η_1 of their fixed laws, then for any event defining the fixed strict margin,

$$\boxed{\widetilde{\Delta} \geq \Delta - \eta_0 - \eta_1.}$$

This is the reverse triangle inequality applied to event probabilities.

For the new capacity-guarded R bridge, exactly the same argument gives

$$\boxed{\widetilde{\Delta}_R^{\text{cap}} \geq \Delta_R^{\text{cap}} - \eta_{R,0} - \eta_{R,1}.}$$

No new perturbation technology is required; only the diagnostic event changes from $\mathcal{A}_{\circ}$ to $\mathcal{A}_{\text{cap}}$.

S12. Finite-sample building block

For a finite alphabet of size m , a fixed $n \geq 1$, independent identically distributed samples with common law P , empirical law $\widehat{P}_n$, and $\alpha \in (0, 1)$, define

$$r_{\text{TV}}(n, m, \alpha) = \min \left\{ 1, \frac{m}{2} \sqrt{\frac{\log(2m/\alpha)}{2n}} \right\}.$$

By Hoeffding's inequality applied to every symbol and a union bound,

$$\boxed{P \left[d_{\text{TV}}(\widehat{P}_n, P) \leq r_{\text{TV}}(n, m, \alpha) \right] \geq 1 - \alpha.}$$

This is conservative; no optimality claim is made.

S13. Whole-envelope simultaneous finite-sample certification

The soundness result bounds the probability of PASS when the quantitative requirements are not met. Acquisition concerns whether the design obtains enough usable observations. A power guarantee concerns the probability of PASS under a specified alternative. Each guarantee below has its own assumptions, and the structural causal conditions require separate assessment.

S13.1 Confidence soundness and acquisition

Fix a finite family of required distributional quantities and a declared parameter domain Θ . Equip the parameter and data spaces with their declared sigma-fields. Require the target $\mathcal{T}$, the coverage and acquisition events, and the implemented PASS event to be measurable. In particular, a set-inclusion decision on a general data space needs its own measurability condition. S19.4 verifies these conditions for the finite-alphabet designs used here. The hard structural conditions are held fixed and must be established by their own valid assessment. For data D , let $\mathcal{C}(D) \subseteq \Theta$ be a simultaneous confidence set and let $\mathcal{T} \subseteq \Theta$ be the quantitative target. Allocate cellwise budgets $\alpha_k > 0$ with $\sum_k \alpha_k \leq \alpha_{\text{infer}}$, and use confidence events valid under the actual sampling design. The quantities may include local margins, handoff mismatch, continuation laws, transcript terms, and support bounds.

Writing $E_\theta = \{\theta \in \mathcal{C}(D)\}$, the union bound gives

$$P_\theta(E_\theta^c) \leq \sum_k \alpha_k \leq \alpha_{\text{infer}}.$$

PASS requires all hard conditions to have been established and $\mathcal{C}(D) \subseteq \mathcal{T}$. If $\theta \notin \mathcal{T}$, PASS implies E_θ^c ; hence

$$P_\theta(\text{PASS} \wedge \theta \notin \mathcal{T}) \leq \alpha_{\text{infer}}.$$

This is quantitative soundness. Any probabilistic hard-condition assessment would require its own error analysis; it is not silently covered by the soft-error budget.

For an adequate-acquisition event A satisfying $P_\theta(A^c) \leq \alpha_{\text{acq}}$,

$$P_\theta(A \cap E_\theta) = 1 - P_\theta(A^c \cup E_\theta^c) \geq 1 - \alpha_{\text{acq}} - \alpha_{\text{infer}}.$$

Neither independence nor a PASS-power interpretation follows. A covered confidence set can intersect both $\mathcal{T}$ and its complement. Mere target membership and successful acquisition therefore do not imply high probability of PASS.

S13.2 A sufficient separation and precision condition for power

Let $\theta = (P_1, \dots, P_K) \in \Theta$, and define

$$\mathcal{T} = \{\theta \in \Theta : g_\ell(\theta) > 0 \text{ for all } \ell = 1, \dots, L\}.$$

Assume that on the relevant parameter domain each target function obeys

$$|g_\ell(P_1, \dots, P_K) - g_\ell(Q_1, \dots, Q_K)| \leq \sum_{k=1}^K L_{\ell k} d_{\text{TV}}(P_k, Q_k), \quad L_{\ell k} \geq 0.$$

These constants are additional quantitative premises for this power statement. Their existence is not asserted for every model, particularly at singular conditional-probability boundaries. Assume the true margins satisfy $g_\ell(\theta) \geq \gamma_\ell > 0$. Let the confidence set be contained in the product of balls centered at $\widehat{P}_k$ with realized radii $\widehat{r}_k$. On A , suppose $\widehat{r}_k \leq r_k$, where the deterministic design bounds satisfy

$$2 \sum_{k=1}^K L_{\ell k} r_k < \gamma_\ell \quad \text{for every } \ell.$$

Also assume that every record in $A \cap E_\theta$ establishes the required hard conditions and meets the remaining declared PASS prerequisites. Then

$$P_\theta(\text{PASS}) \geq 1 - \alpha_{\text{acq}} - \alpha_{\text{infer}}.$$

Proof. On $A \cap E_\theta$, both the true law and any $Q \in \mathcal{C}(D)$ lie in the appropriate balls. For every coordinate,

$$d_{\text{TV}}(P_k, Q_k) \leq d_{\text{TV}}(P_k, \widehat{P}_k) + d_{\text{TV}}(\widehat{P}_k, Q_k) \leq 2\widehat{r}_k \leq 2r_k.$$

Consequently

$$g_\ell(Q) \geq \gamma_\ell - 2 \sum_k L_{\ell k} r_k > 0.$$

Thus $\mathcal{C}(D) \subseteq \mathcal{T}$, and all PASS criteria hold on $A \cap E_\theta$. Apply S13.1 to that event. ■

For a two-arm contrast $\Delta = d_{\text{TV}}(P_1, P_0)$ and retained floor $d_* > 0$, take $g = \Delta - d_*$ and $L_0 = L_1 = 1$. If $\Delta - d_* \geq \gamma > 0$, the sufficient precision requirement is $2(r_0 + r_1) < \gamma$. On coverage, the conservative lower bound $\widehat{\Delta} - r_0 - r_1$ exceeds d_* . The factor two comes from controlling both center error and the width of the confidence set.

These separation values describe a power-qualified alternative or a planning assumption; they are not inferred merely from membership or inserted after observing a favorable result. Without a proved power condition, report only soundness and the actual PASS/INCONCLUSIVE outcome.

S13.3 Conditional cells and random acquisition counts

One explicit admissible design uses a fixed number N of IID pairs (Z_i, Y_i) , with cell tag Z_i defined by a predeclared rule and target conditional law $P_k = \mathcal{L}(Y_i | Z_i = k)$. The cell definition must agree with the scientific target; selection or redefinition of the target after observing responses is not licensed.

Conditional on the complete vector $(Z_1, \dots, Z_N)$, the retained observations in cell k are independent with common law P_k . Their count n_k is then fixed. Thus the S12 radius with budget α_k is valid for each nonempty cell conditionally on that tag vector. Integrating the conditional union bound gives simultaneous coverage at the realized counts. No independence between cell estimates is required by this union bound.

If $n_k = 0$, use the whole probability simplex for that cell and return INCONCLUSIVE when it is required. For a power calculation, predeclare $n_k^{\min} \geq 1$, set

$$A = \bigcap_k \{n_k \geq n_k^{\min}\}, \quad r_k = r_{\text{TV}}(n_k^{\min}, m_k, \alpha_k),$$

and prove an acquisition bound under the declared tag law. Radius monotonicity then gives $\hat{r}_k \leq r_k$ on A . For example, with a fixed N and cell probabilities π_k , a sufficient bound is

$$P(A^c) \leq \sum_k P\{\text{Binomial}(N, \pi_k) < n_k^{\min}\}.$$

Unknown cell probabilities require justified bounds or a separately valid design; they may not be treated as known from favorable observed counts. For multiple fixed-size independent arms the same argument is applied arm by arm. Outcome-dependent stopping, response-based deletion, informative missingness, clustering, and adaptive regime changes require their own valid sampling analysis. Merely substituting the realized count does not resolve these issues.

S13.4 Calibrated sequential designs

A fixed-sample confidence statement applies at its declared deterministic sample size, or under an explicitly valid conditional argument such as S13.3. If the analysis time depends on outcomes, an additional simultaneous or stopping-specific guarantee is required.

For a concrete conservative construction, suppose each cell provides an IID stream with a fixed law, and assign

$$\alpha_{k,n} = \frac{\alpha_k}{n(n+1)}, \quad n \geq 1, \quad \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

Apply S12 at every deterministic within-stream size n , using $r_{k,n} = r_{\text{TV}}(n, m_k, \alpha_{k,n})$. Countable subadditivity gives

$$P\left\{\exists k, n \geq 1 : d_{\text{TV}}(\hat{P}_{k,n}, P_k) > r_{k,n}\right\} \leq \sum_{k,n} \alpha_{k,n} \leq \alpha_{\text{infer}}.$$

Therefore all these intervals cover simultaneously outside one event of probability at most α_{infer} . Evaluation at any finite data-dependent stopping time within the verified streams preserves the same coverage statement. Empty streams remain unassessed. Another valid option is a predeclared finite set of analysis times with per-look budgets whose total is bounded; temporal independence is not required for the union bound.

The statistical guarantee comes from the simultaneous bound or a proved calibration for the particular stopping rule. Preregistration records that valid procedure and prevents post-hoc changes; it does not itself prove optional-stopping validity. The stopping rule, information horizon, cell formation, exclusions, and all error allocations must match the actual protocol. A sequential procedure also needs a separate separation/precision argument if a lower bound on PASS power is claimed.

S14. An explicit finite realization

Part III of this supplement gives one explicit finite model with:

- empty-matching, equal-rate neutral genesis;
- relation-relative DFI-A with $\Delta_A = 1$;
- DFI-H with $\Delta_H = 1$;
- DFI-R(loop) support-loop with $\Delta_R = 1/2$;
- DFI-R(cap) actual-future-capacity effect with $\Delta_R^{\text{cap}} = 1$;
- one global acyclic UCCE atlas with exact handoff/context/kernel/time compatibility;
- UFE zero-error realization.

This proves the refined abstract hypothesis class is nonempty.

The witness is deliberately nonchemical.

S15. Scope of the mathematical results

This supplement does not claim:

- that every printed sufficient block is globally necessary;
- that the DFI-R(loop) support-loop event equals operational-assignment capacity;
- recursive multi-cycle reflexive closure;
- universal finite-size scaling;
- universal parasite immunity;
- universal positive dissipation;
- a chemical or prebiotic realization;
- novelty or priority.

Part II supplies the complete actual-future-capacity certificate. Part III gives the explicit finite construction. Both are included in this supplement.

S16. DFI-R(loop) condition dictionary

This section reproduces the exact condition semantics needed to independently verify the support-loop/re-entry conditions and the finite construction.

S16.1 Variables and interventions

Let

$$A \in \mathcal{A}, \quad S \in \mathcal{S}, \quad A^+ \in \mathcal{A}, \quad W \in \mathcal{W}.$$

Fix

$$a_0 \neq a_1, \quad s_0 \neq s_1.$$

The organization interventions are

$$\boxed{G_i := do(A = a_i)}$$

and the support interventions are

$$\boxed{I_i := do(S = s_i), \quad i = 0, 1.}$$

They are surgical interventions: only the designated structural equation is replaced.

S16.2 R5S† — current support mechanism

There are a fixed exogenous law λ_S , exogenous disturbance E_S , and one structural function

$$\boxed{S = F_S(A, W, E_S).}$$

R5S† requires:

1. no endogenous parent of S outside A, W ; an auxiliary diagnostic output may be pointwise related to S , but it cannot appear as an additional endogenous parent in the fixed support structural equation;
2. no direct certification-protocol path to S outside the declared A, W interface;
3. E_S is exogenous relative to the upstream variables;

4. under G_0, G_1 ,

$$E_S \sim \lambda_S, \quad E_S \perp W;$$

5. the same F_S is used under the allowed current organization interventions and the declared next-cycle re-entry semantics.

Define

$$K_S(s \mid a, w) = P_{\lambda_S}[F_S(a, w, E_S) = s]$$

and, for a context law μ ,

$$\Sigma_{a,\mu}(s) = \sum_w K_S(s \mid a, w) \mu(w).$$

S16.3 R5A† — maintenance mechanism

There are a fixed law λ_A , exogenous disturbance E_A , and

$$A^+ = F_A(S, W, E_A).$$

R5A† requires:

1. no endogenous parent of A^+ outside S, W ;
2. no direct $A \rightarrow A^+$ bypass outside the declared support/context interface;
3. E_A is exogenous relative to the upstream variables;
4. under every G_i, I_i ,

$$E_A \sim \lambda_A, \quad E_A \perp W;$$

5. the same F_A, λ_A are used in all four regimes.

Define

$$K_A(a^+ \mid s, w) = P_{\lambda_A}[F_A(s, w, E_A) = a^+]$$

and

$$M_{s,\mu}(a^+) = \sum_w K_A(a^+ \mid s, w) \mu(w).$$

S16.4 Forward-active support-response class

Fix a reference state a_0 . Define

$$\mathcal{A}_\circ(a_0, \mu) = \{a : \Sigma_{a,\mu} \neq \Sigma_{a_0,\mu}\}.$$

This is a **support-response** class, not an operational-assignment-capacity class.

S16.5 Current-cycle certificate blocks

R0*:

$$P(S = s_i \mid G_i) = 1, \quad i = 0, 1.$$

R1*:

$$P(A^+ \in \mathcal{A}_\circ \mid I_1) > P(A^+ \in \mathcal{A}_\circ \mid I_0).$$

R23* requires one **joint** context law μ under all four regimes:

$$\mathcal{L}(W \mid G_0) = \mathcal{L}(W \mid G_1) = \mathcal{L}(W \mid I_0) = \mathcal{L}(W \mid I_1) = \mu.$$

S16.6 R6† — certification-supported standardized next-support re-entry

Define

$$\mathcal{A}_{\text{cert}} = \text{supp } M_{s_0,\mu} \cup \text{supp } M_{s_1,\mu}.$$

At least one declared next-cycle re-entry regime R must satisfy coverage

$$\boxed{\mathcal{A}_{\text{cert}} \subseteq \text{supp}_R(A^+).$$

For every

$$a \in \text{supp}_R(A^+)$$

the next context must satisfy

$$\boxed{P_R(W_{\text{next}} = w \mid A^+ = a) = \mu(w) \quad \forall w.}$$

The next support-production step uses the **same** F_S . For every $\mu(w) > 0$,

$$\boxed{\mathcal{L}_R(E_{S,\text{next}} \mid A^+ = a, W_{\text{next}} = w) = \lambda_S.}$$

Hence, for every covered state,

$$\boxed{P_R(S_{\text{next}} = s \mid A^+ = a) = \Sigma_{a,\mu}(s).}$$

This identity states the standardized next-support response required by R6 $\dagger$ in the finite construction.

The support-loop theorem establishes the support-loop/re-entry target under

$$R0^* + R1^* + R23^* + R5S^\dagger + R5A^\dagger + R6^\dagger.$$

The support-loop target does not imply actual future operational-assignment capacity; the additional conditions for that conclusion are given in Part II.

S17. UCCE condition dictionary

This section specifies how the local A/H/R certificates can be realized in one causal system. All conditions refer to one fixed common datum, whose components are listed below.

S17.1 One fixed common datum

One theorem invocation uses one fixed datum

$$\boxed{\mathfrak{D}_{\text{UCCE}}}$$

containing:

- fixed A/H/R structural witnesses;
- one global required-regime atlas;
- injective structural embeddings;
- fixed quotient maps q_j ;
- fixed boundary maps ϕ_b ;
- fixed persistent source-regime families;
- fixed continuation-regime families;
- fixed standardized context laws;
- one atlas graph and one joint response-table law;
- one completion-control partition;
- one temporal embedding.

Different sufficient blocks may not choose different witnesses.

S17.2 Persistent and combined state binding

For each boundary $b \in \{AH, HR\}$, let $\mathcal{R}_b^{\text{persist}}$ be the complete persistent source regime family and

$$\mathcal{C}_b^S = \bigcup_{r \in \mathcal{R}_b^{\text{persist}}} \text{supp}_r(q_S(X_S)).$$

Let

$$\phi_b : \mathcal{C}_b^S \rightarrow \mathcal{C}_b^T$$

be a distinction-preserving bijection.

Required-regime binding:

$$P_r[q_T(X_T) = \phi_b(q_S(X_S))] = 1$$

for every

$$r \in \mathcal{R}_b^{\text{persist}}.$$

For every boundary-preserving continuation regime $e \in \mathcal{E}_{j,b}^{\text{cont}}$, the combined regime must also satisfy

$$P_{\text{req}}^{r \oplus e}[q_T(X_T) = \phi_b(q_S(X_S))] = 1.$$

S17.3 Required/combined regime closure

For every

$$b, \quad r \in \mathcal{R}_b^{\text{persist}}, \quad e \in \mathcal{E}_{j,b}^{\text{cont}},$$

the combined surgical regime

$$r \oplus e$$

must belong to the required regime family.

The source part may not directly replace the protected target state/context/ continuation mechanisms, and the continuation part may not replace the protected boundary handoff except as explicitly declared.

S17.4 Fixed and actual continuation laws

For target certificate j , continuation regime e , target quotient class ϑ , context w , and full joint continuation outcome y , let

$$g_{j,e,y}(\vartheta, w)$$

be the reference continuation-response probability specified by the local structural model.

For every relevant target quotient class, the common datum contains a **total predeclared** standardized context law

$$\mu_{j,e,\vartheta}.$$

The fixed target continuation law is

$$\bar{P}_{j,e}^{\vartheta}(\mathbf{Y}_{j,e} = y) = \sum_w g_{j,e,y}(\vartheta, w) \mu_{j,e,\vartheta}(w).$$

For a positive-support source class θ in a combined regime define

$$\nu_{b,r,e,\theta}(w) = P_{\text{req}}^{r \oplus e}(W_j = w \mid q_S(X_S) = \theta).$$

The actual continuation kernel is

$$K_{b,r,e,y}^{\text{act}}(\theta, w) = P_{\text{req}}^{r \oplus e}(\mathbf{Y}_{j,e} = y \mid q_S(X_S) = \theta, W_j = w)$$

on actual positive-support context cells.

S17.5 Bkernel and Bcontext

Kernel invariance requires

$$K_{b,r,e,y}^{\text{act}}(\theta, w) = g_{j,e,y}(\phi_b(\theta), w)$$

for every required positive-support cell and every joint continuation outcome.

Context compatibility requires

$$\sum_w g_{j,e,y}(\phi_b(\theta), w) \nu_{b,r,e,\theta}(w) = \sum_w g_{j,e,y}(\phi_b(\theta), w) \mu_{j,e,\phi_b(\theta)}(w)$$

for every required b, r, e, θ, y .

By the law of total probability these two conditions imply

$$P_{\text{req}}^{r \oplus e}(\mathbf{Y}_{j,e} = y \mid q_S(X_S) = \theta) = \bar{P}_{j,e}^{\phi_b(\theta)}(\mathbf{Y}_{j,e} = y).$$

S17.6 Fixed structural embedding IA-0

Every local A/H/R witness embeds into the same global system while preserving:

1. variable/role injectivity;
2. protected mechanism parent/interface structure;
3. every printed structural/model-class predicate;
4. fixed surgical intervention meaning.

Required-regime law equality alone is not enough.

S17.7 One global intervention atlas

The required regime family contains:

1. every local certificate intervention;
2. every persistent source-side boundary regime;
3. every boundary-preserving continuation regime;
4. every required combined regime.

One acyclic parent graph and one common modular response law generate them.

The intervention block includes:

- IA-0 structural embedding;
- IA-1 local interventional projection;
- IA-2 structural-overlap consistency;
- IA-3 protected-interface preservation;
- IA-4 universal source-faithful boundary semantics;
- IA-5 one global modular response law;
- IA-6 boundary/continuation regime closure;
- support closure.

For completion, the global variable set is partitioned into controlled and free nodes.

Controlled nodes have precommitted total extensions agreeing with every required partial table and preserving all protected IA-0 predicates.

A free node is admissible only when **every** jointly admissible off-domain completion leaves every IA-0 predicate of every fixed witness true.

The extreme choice

$$\boxed{\mathcal{V}_{\text{ctrl}} = \mathcal{V}, \quad \mathcal{V}_{\text{free}} = \emptyset}$$

is allowed. In this case completion safety reduces to direct verification of the single fixed total SCM.

This is the form used by the finite construction in Part III, which instantiates every required regime family and combined cell explicitly.

S17.8 Time compatibility

Every global variable has a strict lexicographic causal/time coordinate (τ, κ) .

Every atlas edge and every $A \rightarrow H \rightarrow R$ handoff must respect the **same** strict order.

In particular zero-delay handoffs require strict within-epoch ordering.

S17.9 Exact target and theorem

CLAIM-AHR* requires one finite acyclic causal system that:

1. contains structurally faithful A/H/R witnesses;
2. realizes their required local laws;
3. preserves every required **joint** continuation law across every positive-support persistent handoff and continuation regime;
4. uses one strict $A \rightarrow H \rightarrow R$ causal order.

The fixed sufficient theorem is:

$$C_A + C_H + C_R + B_{\text{structural}} + B_{\text{state}}^{\text{req}} + B_{\text{state}}^{\text{comb}} + B_{\text{regime}}^{\text{closure}} + B_{\text{kernel}} + B_{\text{context}} + B_{\text{intervention}}^{\text{req}} + B_{\text{time}} \implies \text{CLAIM-AHR}^*.$$

The finite construction in Part III prints the fixed quotient/boundary maps, persistent and continuation families, every combined regime, singleton context laws, response coordinates, structural embedding, completion choice, and time order. It must be checked against those explicit data, not merely against their names.

S18. Explicit witnesses and scope of selected non-implications

S18.1 Heredity and reflexive maintenance are non-equivalent

Both witnesses are conditional on an operational assignment assay; no additional genesis claim is made here. Use the three organization states B, M_0, M_1 of Part III, with challenges $u \in \{0, 1\}$, $\text{read}(M_i, u) = u \oplus i$, and $\text{read}(B, u) = \text{null}$. Whole-organization settings and the fixed challenge assays distinguish M_0, M_1 , and the future-capacity set is $\{M_0, M_1\}$. Orientation labels are diagnostic state labels, not semantic targets in an autonomous rule.

(a) Heredity without support-dependent future capacity

Let a parent organization $P \in \{M_0, M_1\}$ feed $L = P$ and descendant $C = L$. Use constant H context and surgical parent and lineage settings to M_i . Under either kind of setting, the child is M_i ; for event $\{M_1\}$ the probabilities are zero and one. The child equation is stable and has no other parent or noise. Thus H0, H1, H23*, and H5† hold and $\Delta_H = 1$.

At the current maintenance comparison take $P = M_1$. Its challenge-zero output is $O = 1$, and let natural support be $S = O$. The sham preserves $S = 1$, while the block sets $S = 0$ after the same output. Let the actual future assay state be $C^+ = C = L$, unchanged by support. Equivalently the stable maintenance mechanism is $F^+(s, w) = w$ with the same upstream context $w = L = M_1$ in both support arms. Actual future capacity is one in both arms, so its support-induced difference is zero. H therefore holds while positive assignment-reflexive maintenance fails. All supports and future-assay states in this comparison are explicit.

(b) Reflexive maintenance without a separately declared lineage heredity

Take the complete finite R(cap) system of Part III, Sections 7–8, unchanged, including its natural output, distinct support states, stable maintenance mechanism, and actual future cut $C^+ = A_R^+$. Its capacity margin is one. Form its product with an independent fair bit H , and add a declared descendant interface $P_D = A_R$, $L_D = P_D$, $D = M_H$. Permit parent and lineage settings to M_0, M_1 at this interface. The new coordinates have no outgoing arrows into the original R system or its fixed future assay.

Under every such parent or lineage arm, D is uniform on $\{M_0, M_1\}$, so the descendant-channel law has zero contrast. The equation $D = F_D(L_D, H) = M_H$ is stable, with independent noise and constant context, but the positive H1* condition fails. The unchanged R(cap) certificate still holds: marginalizing the independent product coordinates preserves every original interventional law, natural-support condition, and future-assay signature. Consequently R(cap) does not imply heredity at an independently declared lineage interface.

This witness distinguishes maintenance at the declared future assay from orientation transmission to the declared descendant. It does not assert non-heredity at every possible interface, nor does it assert failure of the original Part III H certificate at its different interface. Together (a) and (b) prove Proposition 1 with the stated interfaces. $\square$

S18.2 Elementary witnesses for Table 1

Each example refutes only the implication named in its label. It is not asserted to satisfy an unrelated full A/H/R conjunction. All bit variables take values in $\{0, 1\}$; $\oplus$ is exclusive-or. Undeclared noises are independent and context is constant.

(a) Correlation without operational assignment. Let U be fair and $V = U$, with no effective organization parent. The passive correlation is perfect, but every admissible organization setting has the same challenged response law. The organization-dependence part of operational assignment is absent.

(b) Neutral ensembles and broken physical symmetry. A fair bit Q has an invariant distribution under $q \mapsto 1 - q$, while neither realized value is fixed. If the response is always null for every challenge and both values of Q , this physical symmetry-broken state carries no assignment channel. These are two distinct failures of implication.

(c) Agreement without causal heredity. Let Z be fair and $P = Z, C = Z$. Passive agreement is one, whereas either surgical setting of P leaves C fair. Parent intervention has zero descendant contrast.

(d) Nuisance transmission without assignment heredity. Let a nuisance bit N be copied by $L = N, C = N$, while the declared assignment assay is null in every descendant state and challenge. The nuisance is transmitted perfectly; the assignment-channel diagnostic is constant.

(e) A directed-path cut without a common downstream kernel. Set $P = 0$ at baseline, let Z be independent and fair, and use $L = P \oplus Z, C = L \oplus Z$. Every directed path from P to C passes through L . Under $do(P = i)$, L is fair in both arms but $C = i$. Hence no common kernel T can map the identical two laws of L to these different laws of C . Shared noise between L and C prevents the required screening. This does not contradict Theorem 7, whose common-kernel premise is false.

(f) Generic maintenance without assignment-related support. A system with constant output $O = 1$, support $S = O$, and future viability equal to S maintains itself and is harmed by blocking support. Give it a null assignment assay. Neither current operational assignment nor assignment-related natural support loading follows from this maintenance effect. A reporting coordinate $read(A, u)$ appended to the output while S remains its constant coordinate also fails RASM+.

(g) Covariance without an output effect. Let Z be fair, $O = Z$, and $C^+ = M_1$ when $Z = 1$ and $C^+ = B$ otherwise. The covariance of O with the future-capacity indicator is $1/4$. Setting O alone leaves Z and C^+ unchanged, so the causal effect is zero.

(h) Finite-horizon versus permanent retention. A named orientation survives until the first independent Poisson reset of rate $\lambda > 0$, and each reset erases that named orientation. For any fixed $T > 0$ and $\varepsilon > 0$, choosing λ small enough gives $e^{-\lambda T} \geq 1 - \varepsilon$. Permanent retention has probability zero. This is the elementary special case of Theorem 10, not a stationary or infinite-time retention guarantee.

(i) No class-wide positive error radius without a uniform margin. Take two Bernoulli diagnostic laws with parameters $1/2 - \delta/2$ and $1/2 + \delta/2$, where $0 < \delta < 1$. Their TV separation is δ . Replacing each by Bernoulli($1/2$) costs only $\delta/2$ in each arm and erases the separation. Since δ can approach zero, strict positivity for each individual witness does not imply a common positive robustness radius for the entire class.

(j) No automatic parasite immunity. To any resident finite A/R witness append a parasite count $N \geq 0$ with $N_{t+1} = 2N_t$, with no arrows from N into the resident. Resident certificates are unchanged, while the parasite has growth factor two near zero. Restrictions on parasite coupling, resources, or fitness would be additional hypotheses; no such restriction is part of this non-implication.

(k) Resource inflow without causal maintenance. A positive unused resource inflow can coexist with the stable equation $C^+ = B$, independent of support. Inflow alone gives neither assignment capacity nor its positive support-induced maintenance.

(l) A fixed coarse trajectory without a fine lift. In $(\mathbb{N}, +, 0)$, a zero initial balance, input two, and required output three have no nonnegative residual. The homomorphism $q(n) = \min(n, 2)$ into truncated addition $a \oplus b = \min(a + b, 2)$ maps this same demand to two. The coarse step has residual zero because $2 = 2 \oplus 0$, yet the specified fine step is impossible. A homomorphism alone is not a lifting theorem.

(m) Deletion dependence without global necessity. In the Boolean model class let the target be $C = p \vee q$. Within the architecture $p = 1, q = 0$, dropping p permits $p = q = 0$ and loses the target. Nevertheless $p = 0, q = 1$ satisfies the target outside that architecture. Thus deletion-indispensability in one sufficient presentation does not make p globally necessary.

S18.3 Scope of temporal and intervention-order examples

Let two required events be labelled a, h . Assigned epochs $t_h = 0, t_a = 1$ violate a separate handoff requirement $t_a < t_h$; epoch labels alone do not enforce the intended order. For incompatibility of orders, the temporal constraint $a \prec h$ and an atlas constraint $h \prec a$ are each acyclic, but their union is a directed two-cycle and has no common linear extension. These examples establish failures at the level of ordering data. They do not supply complete independently certified A/H/R models. S17 instead assumes and uses a common acyclic realization and time order; its sufficient implication is unchanged.

S19. Measurable diagnostics and probability domains

S19.1 A sufficient countable-diagnostic condition for operational assignment

Let $(\mathcal{C}, \mathcal{B}_{\mathcal{C}})$ be the causal-state space, let the challenge set A be finite and the response alphabet B finite or countable with its discrete sigma-field. Fix a finite or countable family of admissible organization interventions $(I_j)_{j \in J}$ and the same declared matched comparison. Suppose every baseline and intervened channel coordinate

$$q_{a,b}^0(c) = P_c^{J_a}(Y = b), \quad q_{a,b}^j(c) = P_c^{I_j \cdot J_a}(Y = b)$$

is measurable in c . Here the second expression denotes the declared composed protocol and its matched context, not an arbitrary replacement of its causal semantics. This regularity follows, for example, from measurable intervention maps, readouts and composed probability kernels realizing those very protocols. All interventions in this fixed family are admissible on the stated domain.

The operational-assignment set for this family is

$$E_{\text{OA}} = \bigcup_{a, a' \in A, b \in B} \{c : q_{a,b}^0(c) \neq q_{a',b}^0(c)\} \\ \cap \bigcup_{j \in J, a \in A, b \in B} \{c : q_{a,b}^j(c) \neq q_{a,b}^0(c)\}.$$

Each comparison set is measurable, since it is the inverse image of $\mathbb{R} \setminus \{0\}$ under a difference of measurable real functions. The unions are countable. Equality of probability laws on a countable alphabet is equality of all their point probabilities, so these two factors express precisely the challenge and organization contrasts of the fixed assay. Their intersection is measurable and its indicator is a measurable random variable. Finite or countable Boolean combinations of other declared measurable diagnostic tests remain measurable. If admissibility depends on state, its tests must also be measurable and represented in any signature asserted to determine capacity.

This is a sufficient regularity result, not a reduction of an arbitrary uncountable intervention family to a countable one. For such a larger family, measurability of the existential diagnostic must be supplied as a separate hypothesis or proved from additional regularity. Individual measurable interventions alone do not prove it. $\square$

S19.2 Signatures, equality and capacity preimages

For a finite or countable word family, with a finite or countable response-record alphabet for each word, flatten all response probabilities into a finite or countable coordinate set $\mathcal{J}$. Put the product Borel sigma-field on $[0, 1]^{\mathcal{J}}$, and give the declared signature space its trace sigma-field. Coordinatwise measurable kernels make the signature map measurable: inverse images of coordinate cylinders are measurable and these cylinders generate the product sigma-field. If capacity is expressed by the countable comparisons of S19.1 on these coordinates, its signature subset $\mathfrak{S}_{\text{cap}}^+$ is measurable.

If Ξ and $\widehat{\Xi}$ are measurable maps to this same signature space, then

$$\mathcal{C}_{\text{OA}}^+ = \Xi^{-1}(\mathfrak{S}_{\text{cap}}^+), \quad \mathcal{A}_{\text{cap}} = \widehat{\Xi}^{-1}(\mathfrak{S}_{\text{cap}}^+)$$

are measurable. Signature equality is also measurable: it is the countable intersection of the coordinate-equality events. On this equality event, the two capacity indicators agree pointwise; almost-sure equality therefore transfers their expectations under each specified regime. This proves the probability transfer used by RCapBind+, once the stated binding holds.

For arbitrary signature spaces these same conclusions require the measurable subset and maps explicitly stated in Part II, Section 1, together with a measurable binding event (for example, a measurable diagonal is sufficient). An uncountable product need not have a measurable diagonal in its product sigma-field. Coordinatwise almost-sure equalities over an uncountable family do not replace one simultaneous, measurable binding event. The sufficient countable construction above makes no such extension. $\square$

S19.3 Capacity, composition and finite-error dependencies

For Theorem 9, a common measurable probability kernel $T(h, dc)$ and a measurable future-capacity set E_{cap} give the bounded measurable function $r(h) = T(h, E_{\text{cap}})$. Consequently the two integrals and their

signed difference exist. The usual oscillation bound applies to this same measurable function; a causal graph cut without that common kernel is insufficient.

For DFI-R(cap), the maintenance kernels act on $\mathcal{A}_{\text{cap}}$ only after the preimage conditions in S19.2 or the explicit general-domain premises are met. The fixed support realization and common maintenance mechanism give $\mathcal{L}(A^+ | I_j^+) = M_{s_j, \mu}$. Applying this to the measurable set $\mathcal{A}_{\text{cap}}$, then using the almost-sure indicator equality, gives the printed actual-capacity contrast. RASM+ and all other causal requirements remain separate premises. The capacity-strengthened UCCE target includes these same measurable objects in its common model; gluing the local components does not generate missing measurability.

For UFE, conditioning cells and the correct-binding event G are measurable; the full joint continuation response and transcript are measurable random variables. Conditional laws used on positive-probability branches must be measurable kernels. Each TV defect that is integrated against context must be measurable. For countable response alphabets this follows directly from

$$d_{\text{TV}}(K(w), L(w)) = \frac{1}{2} \sum_b |K(w, \{b\}) - L(w, \{b\})| :$$

the partial sums are measurable and increase to the displayed function. Integration and the mixture/triangle arguments of S8-S11 are then legitimate. Other response spaces require the asserted kernels and defect measurability separately. When $P(G | z) = 0$, the bound one is used, with no conditional kernel claimed on that empty branch. Transport to a capacity diagnostic is evaluation at its fixed measurable event.

In Part III all state, regime, context and response spaces at the certificate cuts are finite discrete spaces. Every diagnostic subset, preimage, equality event and relevant kernel is therefore measurable. In particular, the three-state certificate capacity set remains $\{M_0, M_1\}$; the full seven-matching probe domain has the six nonempty matchings as its OA set. The new domain conditions change neither this distinction nor the four finite margins.

S19.4 Statistical targets and decisions

For the finite-alphabet quantities in S13, use the Borel sigma-field on the declared Borel parameter domain inside the finite product of probability simplexes. The S13.2 coordinatewise TV Lipschitz functions are continuous on this domain, so their finite strict-inequality target $\mathcal{T}$ is measurable. Fixed-event contrasts, finite-alphabet TV distances and the stated confidence radii are measurable functions of their arguments; conditional expressions are used only on their declared positive-support domains.

Under a fixed finite number of trials with finite cell/response alphabets, the data-record space is finite and has its full discrete sigma-field. Thus every specified coverage, adequate-acquisition and PASS decision event, including a decision defined by confidence-set inclusion, is measurable. In the countably indexed sequential design, finite stopped records form a countable set. For a stopping time τ , any event declaring PASS at a finite time is a countable union of measurable events $\{\tau = n, \text{the length-}n \text{ record is accepted}\}$; nontermination is not a finite-time PASS. This establishes measurability for the printed finite-alphabet designs. It does not by itself establish their error rate; S12-S13 supply the distinct sampling and simultaneous-coverage arguments.

For a more general data space or confidence-set construction, the target, each coverage event, acquisition event and implemented PASS event must be measurable by assumption or a separate proof. In particular, arbitrary set-valued confidence regions and an inclusion symbol alone do not ensure a measurable decision. Subject to these conditions, the event inclusion proving soundness and the acquisition/coverage union bound are unchanged. A power claim still needs the printed separation, precision and all-PASS-prerequisite conditions.

S20. Proof of the local discrete-time spectral criterion

This section proves Theorem 12 on a finite-dimensional nonnegative state cone. Let U contain a relative neighborhood of 0 in $\mathbb{R}_+^d$, with finite $d \geq 1$, and let $F : U \rightarrow \mathbb{R}_+^d$ satisfy

$$F(0) = 0, \quad F(y) = Ly + R(y), \quad \frac{\|R(y)\|}{\|y\|} \rightarrow 0 \quad (y \rightarrow 0, y \in U \setminus \{0\}),$$

where L is a $d \times d$ matrix with nonnegative entries. The iteration is $y_{k+1} = F(y_k)$ while defined. Norms are equivalent in this finite dimension, so the little-o condition can be used in each norm below.

S20.1 Subcritical case: an invariant contracting neighborhood

Suppose $\rho(L) < 1$, and choose $\rho(L) < q < 1$. The series

$$w = \sum_{k=0}^{\infty} (L/q)^k \mathbf{1}$$

converges. To justify this finite-matrix step, each Jordan block with eigenvalue of modulus below one has powers that are finite sums of a polynomial in k times a geometric factor of modulus below one; a zero-eigenvalue block is eventually zero. These terms are summable. Conjugating the finitely many blocks proves convergence for L/q . Nonnegativity gives $w_i \geq 1$ and

$$Lw = q(w - \mathbf{1}) \leq qw.$$

Use the weighted maximum norm $\|x\|_w = \max_i |x_i|/w_i$. For any real x ,

$$|Lx| \leq L|x| \leq \|x\|_w Lw, \quad \|Lx\|_w \leq q\|x\|_w.$$

Choose $\epsilon > 0$ with $a = q + \epsilon < 1$. The remainder condition and norm equivalence give a radius $r > 0$ such that the closed cone ball $\{y \geq 0 : \|y\|_w \leq r\}$ lies in U and $\|R(y)\|_w \leq \epsilon\|y\|_w$ throughout that ball. Hence

$$\|F(y)\|_w \leq a\|y\|_w.$$

Because $F(y) \geq 0$, this cone ball is forward invariant. Induction gives a well-defined sequence for every starting point in it and $\|y_k\|_w \leq a^k\|y_0\|_w \rightarrow 0$. This proves local geometric decay without assuming a global dynamical extension. $\square$

S20.2 Supercritical case: a nonnegative first-order direction

Suppose $\rho(L) > 1$. Choose a complex eigenvalue λ of modulus $\rho(L)$ and a nonzero complex eigenvector z . With $v_i = |z_i|$, nonnegativity of L and the coordinatewise triangle inequality give

$$Lv \geq |Lz| = |\lambda z| = \rho(L)v, \quad v \geq 0, \quad v \neq 0.$$

For all sufficiently small $t > 0$, $tv \in U$ and

$$\|F(tv)\|_1 \geq t\|Lv\|_1 - \|R(tv)\|_1 \geq t\rho(L)\|v\|_1 - o(t) > t\|v\|_1.$$

This is the asserted first-order growth direction. It neither says that every nonnegative direction grows nor establishes a global invasion trajectory. In particular, coordinates where $v_i = 0$ need not show strict first-order growth. $\square$

S20.3 Critical case: the linear part does not decide

In one dimension all three maps $F_-(y) = y - y^2$, $F_0(y) = y$, and $F_+(y) = y + y^2$ on a sufficiently small nonnegative neighborhood have $L = 1$ and remainder $o(y)$. For $0 < y_0 < 1$, the first sequence decreases, stays positive and has limit $\ell = \ell - \ell^2$, hence $\ell = 0$. The second is constant. The third increases and exits any fixed sufficiently small positive neighborhood: if it remained bounded there, its positive limit would satisfy $\ell = \ell + \ell^2$, a contradiction. Thus identical critical linearization permits decay, neutrality and local growth. No critical conclusion follows without further nonlinear hypotheses. $\square$

Part II. Actual-future-state capacity-guarded reflexive maintenance

Overview: why actual future-state binding is required

The support-response target $\mathcal{A}_\circ$ and operational-assignment-capacity target $\mathcal{A}_{\text{cap}}$ are distinct. An organization-level classification can indicate capacity even when another coordinate of the actual future state prevents operational assignment. To rule out that mismatch, the certificate requires the specified organization-level signature to agree with the interventional signature of the actual future state. The sham/block comparison uses the natural output-to-support law. Under its stated assumptions, the certificate gives a sufficient route to the one-step maintenance criterion in the main article.

1. Fixed actual future-assay object

Let

$$C^+ \in \mathcal{C}^+$$

be the **actual physical causal state at the fixed actual future operational-assignment assay cut**.
Let

$$\mathcal{W}_{\text{OA}}^+$$

be a fixed separating/generating family of future intervention words used by that operational-assignment assay.

Measurable capacity domain. Use measurable spaces $\mathcal{C}^+$, $\mathcal{A}$, and $\mathfrak{S}_{\text{OA}}^+$. Require the state-indexed response laws to be measurable kernels, Ξ_{OA}^+ and the map $\widehat{\Xi}$ of Section 2 to be measurable, and the fixed subset $\mathfrak{S}_{\text{cap}}^+$ below to be measurable. Require the signature-equality event of Section 3 to be measurable under both comparison regimes. These are domain premises of the theorem in Section 8. They imply measurability of both capacity preimages and their indicators. A countable diagnostic signature with the regularity in S19.1-S19.2 is sufficient; the name “separating/generating” alone is not. For an uncountable assay family the relevant sets and equality event must be established separately.

Define the actual future-interventional signature

$$\Xi_{\text{OA}}^+(c) = (P_c^w)_{w \in \mathcal{W}_{\text{OA}}^+}.$$

Let

$$\mathfrak{S}_{\text{OA}}^+$$

be the signature space and let

$$\mathfrak{S}_{\text{cap}}^+ \subseteq \mathfrak{S}_{\text{OA}}^+$$

be the subset of signatures that satisfy the already-fixed operational-assignment criterion.

Thus the actual future-capacity event is

$$C^+ \in \mathcal{C}_{\text{OA}}^+ \iff \Xi_{\text{OA}}^+(C^+) \in \mathfrak{S}_{\text{cap}}^+.$$

No new semantic criterion is introduced here; this is only the fixed future assay expressed through its future-interventional signature.

2. DFI organization state and capacity realization

Let

$$A \in \mathcal{A}$$

be the current organization state and

$$A^+ \in \mathcal{A}$$

the later DFI organization state.

Fix a **predeclared**, intervention-independent map

$$\widehat{\Xi} : \mathcal{A} \rightarrow \mathfrak{S}_{\text{OA}}^+.$$

Define

$$\mathcal{A}_{\text{cap}} = \{a \in \mathcal{A} : \widehat{\Xi}(a) \in \mathfrak{S}_{\text{cap}}^+\}.$$

This set is diagnostic only.

No physical law, growth law, support law, maintenance law, or intervention selector may read membership in $\mathcal{A}_{\text{cap}}$.

3. RCapBind+ — actual future-state / organization binding

For both matched support comparison regimes I_0^+, I_1^+ , require

$$\boxed{\Xi_{\text{OA}}^+(C^+) = \widehat{\Xi}(A^+) \quad \text{almost surely.}}$$

In particular, A^+ determines every future intervention law in this assay through the same predeclared map $\widehat{\Xi}$. Mere existence of some sufficient function of A^+ is weaker than equality to this fixed map and cannot replace the displayed premise.

In particular,

$$\boxed{\mathbf{1}\{C^+ \in \mathcal{C}_{\text{OA}}^+\} = \mathbf{1}\{A^+ \in \mathcal{A}_{\text{cap}}\} \quad \text{a.s. under } I_0^+, I_1^+ .}$$

Interpretation

RCapBind+ forbids an omitted future coordinate Z^+ that changes operational assignment capacity while leaving A^+ fixed.

The map $\widehat{\Xi}$ is fixed before the support comparison and must be the same under both interventions.

4. Current assignment-mediated output/support path

Let

$$O \in \mathcal{O}$$

be the current physical output of the fixed assignment assay, and let

$$S \in \mathcal{S}$$

be a physical support carrier/state downstream of, or constitutively part of, the declared output-to-future-capacity support path.

Fix the current witness organization

$$a_1 \in \mathcal{A}.$$

RAM-1 — current operational assignment

Require that the current witness organization a_1 , under the declared current causal state realization, **supports the already-fixed current operational-assignment assay**.

This is a pre-existing diagnostic fact about the current witness, not a new capacity definition.

RAM-2 — assignment-mediated output

Require that changing the current assignment-bearing organization through an admissible organization intervention changes the challenge-response law of O under the fixed current assay.

No target mapping, semantic-success score, mutual-information objective, or future-use quantity may enter the output/support law.

RASM+ — assignment-related natural support loading

The fixed-state reflexive certificate additionally requires distinct assignment-bearing organizations a_0, a_1 and two matched natural organization regimes N_j , obtained by admissible $do(A = a_j)$ with the same fixed challenge and otherwise unchanged output/support equations. Only the declared assignment organization is changed: non-target causes of support are represented in the complete matched context, and no unrelated physical support mechanism is surgically altered. This is an intervention-membership hypothesis, not something established by an output contrast alone. The current witness is a_1 . The states $s_0 \neq s_1$ are the same support values as in RB+ below, and

$$P(S = s_j \mid N_j) = 1, \quad \mathcal{L}(W \mid N_j) = \mu, \quad j = 0, 1.$$

R5A†, including its common maintenance function, exhaustive parent interface, noise law and noise/context separation, also holds under these two natural regimes. Thus the natural organization comparison loads distinct support states without an artificial support clamp. This is a separate sufficient hypothesis, not an implication of RAM-2.

To make the relevance explicit, define the diagnostic capacity response

$$q_{\text{cap}}(s, w) = P_{\lambda_A}(F_A(s, w, E_A) \in \mathcal{A}_{\text{cap}}).$$

$R1_{\text{cap}}^+$ then requires $\int [q_{\text{cap}}(s_1, w) - q_{\text{cap}}(s_0, w)] \mu(dw) > 0$. The support states whose natural loading depends on the organization therefore differ in their actual maintenance response, rather than solely in an unused output coordinate. The diagnostic q_{cap} never enters a physical equation or intervention-selection rule. Actual future-state binding remains separately required for the sham/block arms; no unprinted future binding under N0 is assumed.

This fixed-loading version does not claim to characterize all possible stochastic support mediation. The carrier may be a constitutive feature of the same assignment response as O; no independent $O \rightarrow S$ surgical edge is inferred from a pointwise identity.

5. Natural unblocked support regime and matched comparison

Let

$$N_1$$

be the **unblocked natural support regime** with the current witness organization $A = a_1$, the matched context law, and no support-path intervention.

Let the natural output-to-support mechanism be part of the declared physical SCM.

RN+ — natural loading

Require that the current assignment-mediated output under N_1 naturally loads the witness support state s_1 :

$$\boxed{P_{N_1}(S = s_1) = 1}$$

for the fixed-support-state certificate proved here. The sham regime below must also preserve the complete natural $O \rightarrow S$ law rather than replace it. A random-support comparison requires its own joint-law calculation and is not covered by the fixed-state formula merely because the sham preserves the natural law.

RB+ — sham / block semantics

Define two support comparison regimes.

I_1^+ — *sham / normal-support regime*

I_1^+ must preserve the unblocked natural output-to-support law of N_1 .

It may be implemented as a no-op or as a validated sham, but it may not replace S by an artificially favorable state that the assignment-mediated output would not naturally load.

Formally, on the protected current output/support interface,

$$\boxed{\mathcal{L}_{I_1^+}(O, S, W) = \mathcal{L}_{N_1}(O, S, W).}$$

I_0^+ — *matched post-output support block*

I_0^+ differs from I_1^+ only by blocking/clamping the declared **post-output support coupling/state** after the current output O has been generated.

It preserves:

- the current witness organization and challenge;
- the law of O ;
- the matched context/boundary law;
- every upstream mechanism;
- every non-target downstream mechanism;
- gross resource/material conditions except the targeted support coupling.

A destructive global perturbation is not admissible.

For the state-based certificate used below, the two regimes must additionally realize the declared maintenance-interface states

$$\boxed{P(S = s_1 \mid I_1^+) = 1, \quad P(S = s_0 \mid I_0^+) = 1.}$$

For a binary deterministic witness the block may therefore be represented as

$$I_0^+ : do(S = s_0)$$

only after the above natural-sham / post-output-block semantic equivalence has been verified.

6. Maintenance mechanism and matched context

R5A† — intervention-stable maintenance mechanism

Retain the fixed maintenance structure

$$\boxed{A^+ = F_A(S, W, E_A),}$$

with one fixed exogenous law

$$E_A \sim \lambda_A.$$

Require:

1. no endogenous parent of A^+ exists outside S, W ;
2. E_A is exogenous relative to the upstream variables;
3. under both support-comparison arms,

$$E_A \sim \lambda_A, \quad E_A \perp W;$$

4. the same F_A, λ_A are used under I_0^+, I_1^+ .

R23+ — matched complete context law

Define the matched-context condition by

$$\boxed{R23^+ : \quad \mathcal{L}(W \mid I_0^+) = \mathcal{L}(W \mid I_1^+) = \mu.}$$

Thus the complete declared context, not merely selected marginals, is matched between the sham and block arms.

Define

$$K_A(a^+ \mid s, w) = P_{\lambda_A}[F_A(s, w, E_A) = a^+],$$

$$M_{s,\mu}(a^+) = \sum_w K_A(a^+ \mid s, w) \mu(w).$$

If the two support regimes realize the declared maintenance-interface states

$$P(S = s_1 \mid I_1^+) = 1, \quad P(S = s_0 \mid I_0^+) = 1,$$

then

$$\boxed{\mathcal{L}(A^+ \mid I_i^+) = M_{s_i,\mu}.$$

7. R1cap+ — positive future operational-assignment-capacity effect

Require

$$\boxed{M_{s_1,\mu}(\mathcal{A}_{\text{cap}}) > M_{s_0,\mu}(\mathcal{A}_{\text{cap}}).}$$

Define

$$\boxed{\Delta_R^{\text{cap}} = M_{s_1,\mu}(\mathcal{A}_{\text{cap}}) - M_{s_0,\mu}(\mathcal{A}_{\text{cap}}) > 0.}$$

The event $\mathcal{A}_{\text{cap}}$ is diagnostic and fixed from the fixed operational-assignment assay; it does not feed back into dynamics.

8. Main theorem

DFI-R(cap) — Actual-Future-State Capacity-Guarded One-Step Reflexive Maintenance

On the measurable capacity domain of Section 1, assume:

$$RAM-1 + RAM-2 + RASM^+ + RN^+ + RB^+ + RCapBind^+ + R23^+ + R5A^\dagger + R1_{cap}^+.$$

Then

$$\begin{aligned} \rho_{\text{actual}} &= P(C^+ \in \mathcal{C}_{OA}^+ \mid I_1^+) - P(C^+ \in \mathcal{C}_{OA}^+ \mid I_0^+) \\ &= P(A^+ \in \mathcal{A}_{cap} \mid I_1^+) - P(A^+ \in \mathcal{A}_{cap} \mid I_0^+) \\ &= \Delta_R^{\text{cap}} > 0. \end{aligned}$$

Hence the system satisfies the fixed one-step causal reflexive-maintenance criterion: an assignment-mediated output/support route has a positive matched causal effect on **actual future operational-assignment capacity**.

Proof

By the fixed-state realization in RB+, matched context R23+, and intervention-stable maintenance mechanism R5A[†], with its common exogenous law independent of W in both arms,

$$\mathcal{L}(A^+ \mid I_i^+) = M_{s_i, \mu}.$$

Therefore

$$P(A^+ \in \mathcal{A}_{cap} \mid I_i^+) = M_{s_i, \mu}(\mathcal{A}_{cap}).$$

By RCapBind+,

$$\mathbf{1}\{C^+ \in \mathcal{C}_{OA}^+\} = \mathbf{1}\{A^+ \in \mathcal{A}_{cap}\}$$

almost surely under both regimes, so the actual future-capacity probabilities equal the A^+ -capacity probabilities.

Subtracting the two arms gives

$$\rho_{\text{actual}} = \Delta_R^{\text{cap}} > 0.$$

RAM-1 and RAM-2 identify current assignment and organization dependence of the whole output. RASM+ additionally gives $\mathcal{L}(A^+ \mid N_j) = M_{s_j, \mu}$, so the naturally organization-dependent support difference has the positive capacity-response contrast Δ_R^{cap} . This verifies the support-relevant part of the reflexive interpretation. RN+/RB+ identify the comparison as preservation versus matched blocking of the natural post-output support route rather than an artificial favorable state or destructive intervention.

Thus every item of this fixed-state one-step reflexive-maintenance definition is satisfied. The probability identity itself needs only RB+, R23+, R5A[†] and RCapBind+ (and positivity for a positive effect); without RASM+ it need not establish assignment-related support. For example, $O = (\text{read}(A, u), 1)$, $S = 1$ changes its whole-output law under organization interventions, but cannot meet the distinct natural support-loading condition. $\square$

9. A failure without actual-capacity binding

Let

$$A^+ \in \{a_{cap}, a_{non}\}$$

and suppose

$$P(A^+ = a_{cap} \mid I_1^+) = 1, \quad P(A^+ = a_{non} \mid I_0^+) = 1.$$

A naive organization-level capacity margin can therefore be 1.

Now append an omitted actual-future coordinate

$$Z^+ \equiv 0$$

under both arms, and let the fixed OA assay at the actual future cut operate only when

$$Z^+ = 1.$$

Then

$$P(C^+ \in \mathcal{C}_{OA}^+ \mid I_1^+) = P(C^+ \in \mathcal{C}_{OA}^+ \mid I_0^+) = 0.$$

Thus

$$\Delta_R^{\text{cap}}(A^+) > 0 \not\Rightarrow \rho_{\text{actual}} > 0$$

without actual-future-state sufficiency/binding.

RCapBind+ excludes this countermodel. This is an architectural failure example, not a proof that full signature equality is the globally weakest or necessary condition for every capacity-probability bridge. Equality of the two capacity-event indicators almost surely in both arms is already sufficient for the probability transfer alone; other uses of the complete signature may require the stronger condition.

10. Why natural sham semantics is necessary

Suppose the current assignment-mediated output naturally loads

$$S = s_0$$

under the current witness.

If an alleged “sham” intervention instead clamps

$$S = s_1$$

and s_1 happens to produce future capacity, then a positive maintenance contrast can be manufactured even though the natural output never supplies the allegedly supportive state.

RN+/RB+ exclude this artifact by requiring the sham arm to preserve the actual unblocked $O \rightarrow S$ law.

11. Diagnostic quantities and autonomous laws

The sets

$$\mathfrak{S}_{\text{cap}}^+, \quad \mathcal{A}_{\text{cap}}$$

are fixed by the fixed operational-assignment assay before the support comparison.

They are used only to evaluate the causal effect after the physical laws are fixed.

No transition, support law, maintenance law, boundary law, or intervention-selection policy reads:

- target mapping;
- code correctness;
- mutual information;
- semantic success;
- capacity-set membership;
- future usefulness.

The certificate therefore introduces no hidden fitness or foresight.

12. Scope

The certificate proves one-step causal maintenance of **actual future operational-assignment capacity**.

It does not prove:

- assignment genesis;
- causal heredity;
- a second complete maintenance cycle;
- recursive multi-cycle closure;
- indefinite persistence;
- robustness;

- resource admissibility;
- chemistry;
- prebiotic plausibility;
- history;
- novelty or priority.

Part III. Explicit finite non-vacuity construction

Overview

The construction begins with an assignment-neutral, equivariant **genesis stage**. It is followed by an acyclic structural causal model for the post-genesis dynamics. Together, these stages give a finite model that satisfies:

1. neutral-matching decoder-free contingent assignment genesis;
2. relation-relative DFI-A;
3. DFI-H;
4. DFI-R(loop) support-loop/re-entry;
5. DFI-R(cap) actual-future-capacity certificate;
6. exact UCCE composition;
7. UFE with zero implementation error.

The construction proves only that the refined abstract theorem domain is nonempty.

1. Physical assignment states: matchings, not preloaded decoder programs

Let the challenge domain be

$$U = \{0, 1\}$$

and let the **two-element physical matching/output domain** be

$$V_{\text{phys}} = \{0, 1\}.$$

The response alphabet is

$$\boxed{B_{\text{resp}} = V_{\text{phys}} \cup \{\emptyset\},}$$

where $\emptyset$ is a common null response and is **not** a matching endpoint.

An assignment organization is a partial matching between the two two-element physical domains U and V_{phys} .

The origin state is

$$M_{\emptyset} = \emptyset.$$

The two complete matchings are

$$M_0 = \{(0, 0), (1, 1)\},$$

$$M_1 = \{(0, 1), (1, 0)\}.$$

The readout rule is the generic map

$$\begin{aligned} \text{read} : \{\text{partial matchings}\} \times U &\longrightarrow B_{\text{resp}}, \\ \text{read}(M, u) &= \begin{cases} v, & (u, v) \in M, \quad v \in V_{\text{phys}}, \\ \emptyset, & u \text{ unmatched.} \end{cases} \end{aligned}$$

It reads the realized physical link. It does not contain a primitive identity/flip target table.

The symmetry group is

$$\boxed{\Gamma = S_2(U) \times S_2(V_{\text{phys}})}$$

acting naturally on matchings and on the two physical response symbols. The induced action on B_{resp} fixes the common null symbol

$$\gamma(\emptyset) = \emptyset.$$

2. Neutral genesis

Start from $M_\emptyset$. Whenever u and v are unmatched, link (u, v) forms at the same rate

$$\lambda > 0.$$

Links persist during this finite genesis phase.

For $n = 2$, the first link is uniform over four possibilities; the second compatible link is then forced. Therefore

$$P(M_{\text{final}} = M_0) = P(M_{\text{final}} = M_1) = \frac{1}{2}.$$

The process is Γ -equivariant.

At origin, challenge readout is always $\emptyset$, and no permitted nonconstructive origin minor can create a matching. After a complete matching forms, challenge u returns the realized partner; ablating a realized link changes that response.

Thus the construction is the $n = 2$ random-matching witness rather than a symmetric library of preinstalled decoder programs.

Symmetry scope of the witness

The **genesis stage** is assignment-neutral and Γ -equivariant.

The post-genesis A/H/R maintenance SCM below is constructed only to prove joint satisfiability of the stated causal conditions. It is **not** claimed to remain globally Γ -equivariant.

In particular, the later support rule uses the physical response value to define a support state, so the complete post-genesis natural maintenance dynamics may distinguish the two realized orientations.

Therefore this witness proves:

$$\text{assignment-neutral, equivariant genesis} + \text{a post-genesis realization of the stated causal conditions,}$$

not a stronger theorem that the entire A/H/R maintenance dynamics remains assignment-neutral after symmetry breaking.

Genesis domain, certificate-cut domain and assay probes

The genesis/readout domain $\mathcal{M}_{\text{probe}}$ consists of all seven partial matchings. The post-genesis certificate variables $X_A, P_H, L_H, C_H, A_R, A_R^+$ take values in the explicitly restricted domain $\mathcal{M}_{\text{cert}} = \{B, M_0, M_1\}$. Their baseline, required setting/block regimes and maintenance law preserve this domain. This restriction does not describe every intermediate genesis state.

For the post-genesis current and future capacity assay, fix challenges 0 and 1 together with whole-organization ablation to B followed by either challenge. These probes preserve the three-state certificate domain. Their same generic readout/whole-ablation rules also have a total extension to all seven matchings; $\widehat{\Xi}$ is the restriction of that extension to $\mathcal{M}_{\text{cert}}$. On the full probe domain, all six nonempty matchings support OA: a single edge gives a linked-versus-unmatched challenge contrast, and whole-organization ablation changes the response. Thus $\{M_0, M_1\}$ is the capacity set only on the declared certificate-cut domain, not on all seven matching states. Partial-edge origin diagnostics belong to the genesis domain; they are not silently added as post-genesis handoff regimes.

3. Global finite SCM after the genesis cut

Let

$$M_{\text{final}} \in \{M_0, M_1\}$$

be the **physical complete matching produced by the genesis process**. It is not represented by an additional latent decoder coordinate.

Use variables in strict causal order:

$$X_A \prec V_A \prec P_H \prec L_H \prec C_H \prec A_R \prec O_R \prec S_R \prec A_R^+ \prec S_{\text{next}}.$$

All context variables are deterministic constants.

The baseline equations are

$$U_A := 0,$$

with domain $U_A \in \{0, 1\}$; a challenge intervention replaces this equation surgically. Also set

$$\begin{aligned} X_A &:= M_{\text{final}}, \\ V_A &:= \text{read}(X_A, U_A), \\ P_H &:= X_A, \quad L_H := P_H, \quad C_H := L_H, \quad A_R := C_H. \end{aligned}$$

For the R block, define the diagnostic output

$$O_R := \text{read}(A_R, 0).$$

Use the explicit non-capacity reference organization

$$\boxed{B := M_\emptyset.}$$

Hence

$$\text{read}(B, u) = \emptyset.$$

Define the fixed support structural function directly on the organization state:

$$\boxed{F_S(a) := \mathbf{1}\{\text{read}(a, 0) = 1\}.}$$

The current support state is

$$\boxed{S_R := F_S(A_R).}$$

Pointwise,

$$S_R = \mathbf{1}\{O_R = 1\},$$

but O_R is not an additional endogenous parent of S_R in the fixed R5S $\dagger$ certificate. Rather, O_R is the diagnostic component of the same assignment-mediated physical response, while S_R is the support-relevant constitutive indicator generated directly from A_R .

Thus the exact fixed support interface is

$$S_R = F_S(A_R, W_R, E_S)$$

with singleton context W_R and degenerate support noise E_S .

Let

$$Z \sim \text{Bernoulli}(1/2)$$

be independent fresh maintenance noise and define

$$A_R^+ = \begin{cases} B, & S_R = 0, \\ M_Z, & S_R = 1. \end{cases}$$

Finally,

$$\boxed{S_{\text{next}} = F_S(A_R^+) = \mathbf{1}\{\text{read}(A_R^+, 0) = 1\}.}$$

Hence

$$S_{\text{next}}(B) = 0, \quad S_{\text{next}}(M_0) = 0, \quad S_{\text{next}}(M_1) = 1,$$

which is exactly the same support-response rule used in the current cycle.

Conditional on the normal support state $S_R = 1$, maintenance redraws the two named orientations uniformly. This does **not** imply global post-genesis orientation neutrality, because the natural support state itself depends on the realized assignment-mediated output.

4. Operational-assignment capacity

On $\mathcal{M}_{\text{cert}}$, the diagnostic capacity set is

$$\boxed{\mathcal{A}_{\text{cap}} = \{M_0, M_1\}.}$$

The reference state B is not assignment-capable.

This set is not read by any physical equation.

5. DFI-A

Use the DFI-A ontology

$$\{\perp, M_0, M_1\}, \quad \perp = M_\emptyset.$$

The nuisance D is constant and output noise is degenerate.

The output structural law is the same generic matching readout

$$V_A = \text{read}(X_A, U_A).$$

Use later certification protocols J_i whose sole surgical action on the declared endogenous system is

$$\text{do}(X_A = M_i), \quad i = 0, 1.$$

Equivalently, J_i is an admissible organization-setting intervention label, not an extra causal parent of V_A . The structural equation for V_A contains no J , and the same readout law remains valid under counterfactual interventions that decouple J from X_A .

Use witness input

$$U_A = 0.$$

Then

$$V_A = 0 \text{ under } J_0, \quad V_A = 1 \text{ under } J_1.$$

Hence C1* holds. C2* holds because nuisance is identical. C0 holds because the origin matching is empty.

M1 is satisfied in the declared finite model class: every decoder-relevant internal organization for the certified relation is the explicit matching state; there is no latent decoder variable.

M3† holds because V_A has no endogenous parent outside U_A, X_A, D , the noise is exogenous/degenerate, and the same matching-readout function is used in every regime.

Thus

$$\boxed{\Delta_A = 1.}$$

The stronger decoder-free-genesis conclusion comes from Part III, Sections 1–2, not from DFI-A alone.

6. DFI-H

Take

$$P = P_H, \quad L = L_H, \quad C = C_H,$$

with witness states

$$p_i = l_i = M_i.$$

The complete H context is constant.

Under

$$G_i^H = \text{do}(P_H = M_i),$$

the equation $L_H = P_H$ gives H0*.

Under

$$I_i^H = \text{do}(L_H = M_i),$$

the child equation $C_H = L_H$ gives different child laws. For event

$$B_H = \{M_1\},$$

the probabilities are 0 and 1. Hence H1*.

H23* holds by constant joint context.

H5† holds because $C_H = L_H$ has no parent outside the declared lineage/context interface, uses no nontrivial noise, and is intervention-stable.

Therefore

$$\boxed{\Delta_H = 1.}$$

7. Fixed DFI-R support-loop/re-entry

Use the two **same assignment classes handed off by H** as the local R witness:

$$a_0 = M_0, \quad a_1 = M_1,$$

with support states

$$s_0 = 0, \quad s_1 = 1.$$

Under

$$G_0^R = do(A_R = M_0)$$

the output challenge 0 gives $O_R = 0$, hence $S_R = 0$.

Under

$$G_1^R = do(A_R = M_1)$$

the same challenge gives $O_R = 1$, hence $S_R = 1$.

Thus $R0^*$ holds.

The context-averaged support laws are

$$\Sigma_{M_0, \mu} = \delta_0, \quad \Sigma_{M_1, \mu} = \delta_1, \quad \Sigma_{B, \mu} = \delta_0.$$

Therefore, relative to M_0 ,

$$\boxed{\mathcal{A}_\cup(M_0, \mu) = \{M_1\}.$$

Under

$$I_0^R = do(S_R = 0),$$

the maintenance law gives

$$A_R^+ = B.$$

Under

$$I_1^R = do(S_R = 1),$$

the later organization is uniformly M_0 or M_1 .

Hence

$$M_{1, \mu}(\mathcal{A}_\cup) = \frac{1}{2}, \quad M_{0, \mu}(\mathcal{A}_\cup) = 0,$$

so

$$\boxed{\Delta_R = \frac{1}{2} > 0.}$$

$R23^*$ holds by constant context. $R5S^\dagger$ now holds **structurally** because the support equation is exactly $S_R = F_S(A_R)$ with no endogenous parent outside the declared A_R/W_R interface. $R5A^\dagger$ holds by the displayed maintenance parent set, fresh independent maintenance noise, and intervention stability.

For $R6^\dagger$ use a declared re-entry regime whose A_R^+ -support contains

$$\{B, M_0, M_1\},$$

with the same constant next context, the same already-declared support structural function F_S , and degenerate unchanged support noise.

Therefore

$$\begin{aligned} P_R(S_{\text{next}} = 1 \mid A_R^+ = B) &= 0, \\ P_R(S_{\text{next}} = 1 \mid A_R^+ = M_0) &= 0, \\ P_R(S_{\text{next}} = 1 \mid A_R^+ = M_1) &= 1. \end{aligned}$$

These are exactly

$$\Sigma_{B, \mu}, \quad \Sigma_{M_0, \mu}, \quad \Sigma_{M_1, \mu}.$$

Thus the fixed $R6^\dagger$ next-support identity holds on every certification-relevant state, and fixed DFI-R(loop) is satisfied using the same M_0/M_1 boundary classes supplied by the H block.

8. DFI-R(cap) actual-future-capacity certificate

The future assay cut is represented explicitly.

Let

$$\boxed{C^+ := A_R^+}$$

The future assay state has no additional coordinate beyond the variables specified in this construction.

Let the fixed future OA intervention words be the same challenge/organization assays used to test whether a later matching supports operational assignment. The future-interventional signature map is therefore

$$\Xi_{\text{OA}}^+(C^+) = \widehat{\Xi}(A_R^+)$$

identically, with

$$\mathcal{A}_{\text{cap}} = \{M_0, M_1\}.$$

Hence the actual-future-state binding condition holds pointwise:

$$\boxed{\mathbf{1}\{C^+ \in \mathcal{C}_{\text{OA}}^+\} = \mathbf{1}\{A_R^+ \in \mathcal{A}_{\text{cap}}\}.$$

Natural sham / matched block semantics

Use current witness organization

$$A_R = M_1.$$

Under the unblocked natural regime N_1 ,

$$O_R = \text{read}(M_1, 0) = 1,$$

so the natural support rule loads

$$S_R = 1.$$

Define I_1^+ as a sham/no-op on the output-to-support interface. It preserves

$$(O_R, S_R) = (1, 1)$$

and the same constant context.

Define I_0^+ as a **post-output** support block: it leaves

$$A_R = M_1, \quad O_R = 1$$

unchanged and clamps only the downstream support state/coupling to

$$S_R = 0.$$

No other structural equation or context variable is changed.

Thus the sham arm does not artificially substitute a favorable support state; it preserves the state naturally loaded by the current assignment-mediated output.

Capacity effect

Under I_1^+ , the maintenance law gives

$$A_R^+ \in \{M_0, M_1\}$$

with equal probabilities, so

$$P(A_R^+ \in \mathcal{A}_{\text{cap}} \mid I_1^+) = 1.$$

Under I_0^+ ,

$$A_R^+ = B,$$

so

$$P(A_R^+ \in \mathcal{A}_{\text{cap}} \mid I_0^+) = 0.$$

Therefore

$$\boxed{\Delta_R^{\text{cap}} = 1}$$

and, by the exact future-state binding,

$$\boxed{P(C^+ \in \mathcal{C}_{\text{OA}}^+ \mid I_1^+) - P(C^+ \in \mathcal{C}_{\text{OA}}^+ \mid I_0^+) = 1.}$$

For RASM+, choose $N_j = do(A_R = M_j)$ with the printed unchanged support and maintenance functions. Then $S_R = F_S(M_j) = j$ almost surely, contexts are constant, and the same independent fair maintenance noise applies. These are already allowed organization-setting regimes. The support is a constitutive response feature as specified in Section 3, not a new parent edge from O_R . Hence the toy also meets the explicit assignment-related support condition and satisfies the fixed-state capacity-guarded reflexive-maintenance certificate.

This is an explicit one-step future-capacity effect only. No second complete maintenance cycle or global post-genesis Γ -equivariance is claimed.

9. UCCE exact composition — explicit fixed datum

This section gives the complete specification of one fixed UCCE datum. The subsections give its state identifications, regime families, continuation and context laws, intervention atlas, and causal order.

9.1 Global variables and total SCM

Include the origin variable

$$A_0 := M_\emptyset$$

and the post-genesis variables

$$X_A, U_A, V_A, P_H, L_H, C_H, A_R, O_R, S_R, A_R^+, S_{\text{next}}.$$

The genesis process supplies

$$P(X_A = M_0) = P(X_A = M_1) = \frac{1}{2}.$$

No additional orientation variable is introduced in the post-genesis SCM: X_A is the physical matching itself.

All structural functions printed in Part III, Sections 3–8 are total on their declared finite domains. Use the strict order

$$A_0 \prec X_A \prec U_A \prec V_A \prec P_H \prec L_H \prec C_H \prec A_R \prec O_R \prec S_R \prec A_R^+ \prec S_{\text{next}}.$$

9.2 Quotient classes and boundary maps

Let

$$\Theta := \{0, 1\}.$$

On the two positive certification classes M_0, M_1 define

$$q_A(M_i) = i, \quad q_H^{\text{in}}(M_i) = i, \quad q_H^{\text{out}}(M_i) = i, \quad q_R(M_i) = i.$$

For A/H,

$$\boxed{\phi_{AH}(i) = i.}$$

For H/R,

$$\boxed{\phi_{HR}(i) = i.}$$

Both boundary maps are fixed distinction-preserving bijections.

9.3 Persistent source-regime families

For A/H define

$$\mathcal{R}_{AH}^{\text{persist}} = \{r_A^*, r_A^0, r_A^1\},$$

where

$$r_A^* := \text{post-genesis baseline}, \quad r_A^i := do(X_A = M_i).$$

For H/R define

$$\mathcal{R}_{HR}^{\text{persist}} = \{r_H^*, r_H^{P,0}, r_H^{P,1}, r_H^{L,0}, r_H^{L,1}\},$$

with

$$r_H^* := \text{baseline},$$

$$r_H^{P,i} := do(P_H = M_i), \quad r_H^{L,i} := do(L_H = M_i).$$

Every positive-support source class under these regimes lies in Θ .

9.4 Boundary-preserving continuation families

For H after the A/H boundary define

$$\mathcal{E}_{H,AH}^{\text{cont}} = \{e_H^*, e_H^0, e_H^1\},$$

where

$$e_H^* := \text{no target intervention}, \quad e_H^j := do(L_H = M_j).$$

For R after the H/R boundary define

$$\mathcal{E}_{R,HR}^{\text{cont}} = \{e_R^*, e_R^0, e_R^1\},$$

where

$$e_R^* := \text{no support intervention}, \quad e_R^k := do(S_R = k).$$

Every continuation intervention acts strictly downstream of the corresponding boundary state.

9.5 Combined-regime closure and state binding

Define the complete A/H combined family by

$$\mathcal{R}_{AH}^{\text{comb}} = \{r \oplus e : r \in \mathcal{R}_{AH}^{\text{persist}}, e \in \mathcal{E}_{H,AH}^{\text{cont}}\}.$$

It contains exactly

$$3 \times 3 = 9$$

regimes.

Define the complete H/R combined family by

$$\mathcal{R}_{HR}^{\text{comb}} = \{r \oplus e : r \in \mathcal{R}_{HR}^{\text{persist}}, e \in \mathcal{E}_{R,HR}^{\text{cont}}\}.$$

It contains exactly

$$5 \times 3 = 15$$

regimes.

Every member of both sets belongs to the required atlas.

Because the continuation interventions act strictly downstream,

$$q_H^{\text{in}}(P_H) = \phi_{AH}(q_A(X_A))$$

holds almost surely under every A/H persistent and combined regime, and

$$q_R(A_R) = \phi_{HR}(q_H^{\text{out}}(C_H))$$

holds almost surely under every H/R persistent and combined regime.

9.6 Local required regimes in the same atlas

The same finite atlas also contains:

DFI-A

$$j_A^i := do(U_A = 0, X_A = M_i), \quad i = 0, 1.$$

DFI-H

$$G_H^i := do(P_H = M_i), \quad I_H^i := do(L_H = M_i).$$

DFI-R(loop)

$$G_R^i := do(A_R = M_i), \quad I_R^i := do(S_R = i).$$

For $R6^\dagger$ define a precommitted randomized re-entry regime R_{re} . Introduce an exogenous intervention coin

$$C_{\text{re}} \sim \text{Bernoulli}(1/2)$$

independent of every model noise and context, and surgically replace only the current support equation by

$$S_R := C_{\text{re}}.$$

The maintenance equation $A_R^+ = F_A(S_R, w_R, Z)$ remains unchanged.

Therefore

$$P_{R_{\text{re}}}(A_R^+ = B) = \frac{1}{2},$$

$$P_{R_{\text{re}}}(A_R^+ = M_0) = P_{R_{\text{re}}}(A_R^+ = M_1) = \frac{1}{4},$$

so the re-entry regime has positive A_R^+ -support on

$$\{B, M_0, M_1\}.$$

The next context is the same singleton w_R , and the next support is computed by the same function

$$S_{\text{next}} = F_S(A_R^+).$$

Thus R6† coverage is achieved without directly preparing A_R^+ or changing the maintenance mechanism.

DFI-R(cap) The natural current-witness regime is

$$N_1 := do(A_R = M_1)$$

with no support intervention.

The sham arm I_1^+ preserves this protected output/support law.

The block arm is

$$I_0^+ := do(A_R = M_1, S_R = 0),$$

with the support intervention applied after the diagnostic output O_R is generated.

9.6A Explicit finite required-regime atlas

Let $\mathcal{J}_{\text{local}}$ be the finite set of local DFI-A/H/R regimes printed in Section 9.6, including the R6† re-entry regime and the current capacity-certificate sham/block regimes.

Define

$$\begin{aligned} \mathcal{J}_{\text{toy}}^{\text{req}} = \mathcal{J}_{\text{local}} \\ \cup \mathcal{R}_{AH}^{\text{persist}} \cup \mathcal{E}_{H,AH}^{\text{cont}} \cup \mathcal{R}_{AH}^{\text{comb}} \\ \cup \mathcal{R}_{HR}^{\text{persist}} \cup \mathcal{E}_{R,HR}^{\text{cont}} \cup \mathcal{R}_{HR}^{\text{comb}}. \end{aligned}$$

No required regime is supplied by an existential placeholder outside this finite set.

9.7 Fixed context laws

Let

$$W_H = \{w_H\}, \quad W_R = \{w_R\}.$$

For every target class and every continuation regime set

$$\mu_{H,e,\vartheta} = \delta_{w_H}, \quad \mu_{R,e,\vartheta} = \delta_{w_R}.$$

Under every combined regime and every positive-support source class,

$$\nu_{AH,r,e,\theta} = \delta_{w_H}, \quad \nu_{HR,r,e,\theta} = \delta_{w_R}.$$

9.8 Fixed H continuation coordinates

Use

$$\mathbf{Y}_{H,e} := (L_H, C_H).$$

For target class ϑ ,

$$y_H^{\epsilon_H}(\vartheta) = (M_\vartheta, M_\vartheta),$$

and for $j \in \Theta$,

$$y_H^{e_j^j}(\vartheta) = (M_j, M_j).$$

Define

$$g_{H,e,y}(\vartheta, w_H) = \mathbf{1}\{y = y_H^e(\vartheta)\}.$$

The actual H structural equations under every combined regime give the same deterministic outcome. Hence

$$K_{AH,r,e,y}^{\text{act}}(\theta, w_H) = g_{H,e,y}(\phi_{AH}(\theta), w_H).$$

9.9 Fixed R continuation coordinates

Use the joint outcome

$$\mathbf{Y}_{R,e} := (S_R, A_R^+, S_{\text{next}}).$$

For target class $\vartheta = 0$, under e_R^* ,

$$\mathbf{Y}_{R,e_R^*} = (0, B, 0)$$

almost surely.

For target class $\vartheta = 1$, under e_R^* ,

$$\begin{aligned} P[(S_R, A_R^+, S_{\text{next}}) = (1, M_0, 0)] &= \frac{1}{2}, \\ P[(S_R, A_R^+, S_{\text{next}}) = (1, M_1, 1)] &= \frac{1}{2}. \end{aligned}$$

Under $e_R^0 = do(S_R = 0)$, for either target class,

$$\mathbf{Y}_{R,e_R^0} = (0, B, 0)$$

almost surely.

Under $e_R^1 = do(S_R = 1)$, for either target class,

$$\begin{aligned} P[(S_R, A_R^+, S_{\text{next}}) = (1, M_0, 0)] &= \frac{1}{2}, \\ P[(S_R, A_R^+, S_{\text{next}}) = (1, M_1, 1)] &= \frac{1}{2}. \end{aligned}$$

Define

$$g_{R,e,y}(\vartheta, w_R)$$

to be these fixed probabilities.

Because the global SCM uses the same F_S, F_A and the same $Z \sim \text{Bernoulli}(1/2)$ in every required regime,

$$K_{HR,r,e,y}^{\text{act}}(\theta, w_R) = g_{R,e,y}(\phi_{HR}(\theta), w_R).$$

9.10 Explicit context compatibility

For every required boundary, regime, source class, and joint continuation outcome,

$$\begin{aligned} \sum_w g_{j,e,y}(\phi_b(\theta), w) \nu_{b,r,e,\theta}(w) \\ &= g_{j,e,y}(\phi_b(\theta), w_j) \\ &= \sum_w g_{j,e,y}(\phi_b(\theta), w) \mu_{j,e,\phi_b(\theta)}(w). \end{aligned}$$

Therefore

$$B_{\text{context}} = 1$$

and Sections 9.8–9.9 give

$$B_{\text{kernel}} = 1.$$

9.11 Structural embedding

Every local certificate mechanism is literally a protected submechanism of the same global SCM:

$$\begin{aligned}
V_A &= \text{read}(X_A, U_A), \\
L_H &= P_H, \quad C_H = L_H, \\
\boxed{S_R &= F_S(A_R)}
\end{aligned}$$

with no endogenous parent outside the declared A_R/W_R interface,

$$A_R^+ = F_A(S_R, w_R, Z),$$

and

$$S_{\text{next}} = F_S(A_R^+).$$

Thus the fixed parent/interface predicates, intervention semantics, and exogenous-noise conditions are preserved structurally.

Hence

$$\boxed{B_{\text{structural}} = 1.}$$

9.12 One completion-closed global atlas

Use the full finite SCM as the global response-function atlas and set

$$\boxed{\mathcal{V}_{\text{ctrl}} = \mathcal{V}, \quad \mathcal{V}_{\text{free}} = \emptyset.}$$

Every structural function is total and precommitted on its finite parent domain. Every required local, persistent, continuation, and combined regime is a modular surgical replacement of one of those equations.

Because there are no free nodes, no off-support completion choice remains.

Thus IA-0 through IA-6, support closure, and controlled completion hold by direct inspection, giving

$$\boxed{B_{\text{intervention}}^{\text{req}} = 1.}$$

9.13 Common causal/time order

Use the strict order from Section 9.1.

Every protected structural edge and every $A \rightarrow H \rightarrow R$ handoff respects this same order. The support block acts after O_R in the declared within-epoch order and before A_R^+ .

Therefore

$$\boxed{B_{\text{time}} = 1.}$$

9.14 One fixed UCCE datum

Define

$$\boxed{\mathfrak{D}_{\text{toy}}}$$

to be exactly the quotient maps, boundary maps, persistent families, continuation families, all combined regimes, singleton context laws, response coordinates g , actual context laws ν , structural embeddings, completion choice, and temporal order printed in Sections 9.1–9.13.

This is one fixed datum for the entire proof.

Consequently every printed UCCE sufficient obligation is instantiated in one finite model and

$$\boxed{\text{CLAIM-AHR}^*}$$

holds at the fixed local-certificate semantics.

Adding DFI-R(cap) yields the strengthened interpretation in which the R component also satisfies actual future-assignment capacity.

10. UFE exact-zero case

Take the intended mathematical model and realized implementation to be exactly this same SCM.

For every required handoff cell,

$$s_z = k_z = c_z = \iota_z = 0.$$

Therefore every UFE transport theorem gives

$$\sigma_z = \beta_z = 0.$$

The local margins are

$$\Delta_A = 1, \quad \Delta_H = 1, \quad \Delta_R = \frac{1}{2}, \quad \Delta_R^{\text{cap}} = 1.$$

All hard structural/surgical/atlas/order conditions hold exactly. Retained floors for Δ_A , Δ_H , and Δ_R^{cap} may be any positive values strictly below one. If the support-loop margin Δ_R is also required, its retained floor must instead be strictly below $1/2$.

Thus the zero-error special case of the fixed UFE envelope holds.

11. Non-vacuity conclusion

the refined abstract DFI-A/H + DFI-R(loop) support-loop + DFI-R(cap) + UCCE + UFE hypothesis class is nonempty.

The same finite model includes a neutral-matching genesis stage.

This proves only abstract mathematical non-vacuity. It does not prove chemistry, prebiotic plausibility, early-Earth history, recursive multi-cycle closure, indefinite persistence, or novelty.

References

Reference numbering follows the main article.

31. Abramsky, S., and Brandenburger, A. “The Sheaf-Theoretic Structure of Non-Locality and Contextuality.” *New Journal of Physics* 13:113036 (2011). doi:10.1088/1367-2630/13/11/113036.